Forecasting Firm-Level Tail Risk with Structured Mixture Density Networks*

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Abstract

This paper develops structured mixture density networks for forecasting stock return distributions. Macroeconomic inputs determine component probabilities shared by all firms in a month. Firm characteristics determine returns within each component, with extensions that allow macroeconomic conditions to adjust firm features. Across 216 out-of-sample formation months from 2007 to 2024, all three structured specifications improve density, distribution, and tail forecast scores relative to neural Normal and unrestricted mixture benchmarks. Their extreme tail-risk stock selections also overlap much more with one another than with those of the fused benchmarks. The models also provide an exact decomposition of expected shortfall (ES). After controlling for total ES, historical exposures, and predicted mean and volatility, the low-mean component contribution distinguishes stocks with different crash risks in bad markets. In the bottom quintile of market excess returns, the lowest and highest residual contribution deciles differ in crash rates by 6.4–7.1 percentage points using 5% ES and 6.2–7.6 points using 10% ES, with small differences in other months. Two additional random seeds and fixed-epoch training preserve the direction of these findings. ES-sorted portfolios also separate stocks with different losses during market declines. The evidence supports a structured forecasting design with common monthly probabilities and an informative decomposition of firm-level tail risk.

Keywords: expected shortfall, density forecasting, mixture density networks, downside exposure, macroeconomic conditioning.
JEL codes: C45, C53, C58, G11, G12.

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1 Introduction

Large stock losses matter for portfolio choice and risk management. Forecasting them requires both a description of the market environment and a way to distinguish firms within that environment. This paper gives these two tasks separate roles in a return-distribution model: macroeconomic information determines state probabilities shared by all firms, while firm characteristics determine each firm's returns within a state. I study whether this structure improves forecasts and whether the resulting decomposition of tail risk identifies stocks that suffer larger losses when the market falls.

The structure addresses two distinct questions. The first is how to use common information. Macroeconomic variables have one observation per month, whereas firm characteristics vary across thousands of stocks. A shared probability network uses that common information to describe the same monthly environment for every firm. The firm network uses each stock's characteristics to produce different return distributions within each state. The second question is what a tail-risk forecast contains. Two stocks can have similar expected shortfall (ES), the average predicted return in the lower tail, while different mixture components account for their losses. The model reports this composition for each forecast and allows its economic information to be tested.

I use a structured mixture density network (MDN) to separate these roles. An MDN predicts the parameters of a return distribution, rather than only the expected return (Bishop, 1994). In the baseline Structured K3 model, macro-financial variables determine three component probabilities. Every firm shares these probabilities within a month. Firm characteristics and a learned industry embedding determine each firm's component means and standard deviations. The embedding represents an industry with a small set of estimated features. The probabilities describe the common environment. The component parameters describe how outcomes differ across firms within that environment.

Structured K3 is the baseline. Two extensions allow macroeconomic conditions to adjust the features learned from firm characteristics through feature-wise linear modulation (FiLM). FiLM Sigma applies the adjustment to component standard deviations; FiLM All applies it to both means and standard deviations. All three specifications retain the same organizing principle: firms share monthly state probabilities and have different returns within each state. Comparing the three shows whether the findings persist when macroeconomic information has a broader role in the firm-level distributions.

I compare the structured models with a historical Normal forecast, linear quantile regression, a neural Normal model, and an unrestricted three-component neural mixture. Each model uses information from formation month t to forecast the raw stock return in the

next calendar month. The models are estimated in annual rolling windows with 72 training months, 12 validation months, and 12 test months. Formation dates run from January 2007 through December 2024. The common sample contains 480,446 stock–months, with returns observed from February 2007 through January 2025.

The structured models improve forecast accuracy as a group. Their continuous ranked probability score (CRPS), which evaluates the full predictive distribution, ranges from 0.07123 to 0.07222, compared with 0.07803 for Fused K1 and 0.08132 for Fused K3. Their 5% joint value-at-risk (VaR)–expected-shortfall scores range from -1.3701 to -1.3516, compared with -1.2173 and -1.1774 for the two fused benchmarks. At 10%, the structured scores range from -1.5653 to -1.5489, compared with -1.4077 and -1.3606. Paired monthly comparisons support the gains for all three structured specifications. The gains hold across all three variants, supporting a forecasting design that combines shared monthly probabilities with common-state training.

The stock selections show that the three structured variants identify a similar set of stocks. Across the 5% and 10% ES levels, their average extreme-decile overlaps are about 88%–94%, compared with 67%–78% between structured and fused models. Thus, changing how macroeconomic information enters the firm-level parameters produces much smaller selection differences than changing from the structured design to a fused model. Section 3.4 defines the overlap measure and reports both extremes.

Within the forecast evaluation, I compare each mixture with a Normal distribution that has exactly the same predicted mean and variance. All three mixtures improve full-sample log scores and forecasts of losses beyond a fixed cutoff. These gains show that distribution shape provides useful information beyond the first two moments.

I next examine the economic information in the forecasts. Each month, I form portfolios using that month's ES predictions and then measure their subsequent returns. An equal-weighted position that buys the highest-ES stocks and sells the lowest-ES stocks earns about 6% per bad market month at both ES levels. Stocks selected for less severe predicted tail losses therefore perform better relative to stocks selected for more severe losses when the market falls. Equal-weighted gross alphas are also positive and significant under all five factor models considered.

I then study the sources of the predicted tail loss. Expected shortfall (ES) is the average predicted return in the lower tail. I keep ES in signed return units, so a more negative value means a worse loss. Each mixture component contributes its share of lower-tail outcomes multiplied by its average return in that tail. The contributions add up exactly to total ES. For each test window, I label the components using their mean rankings in the training and validation sample. The label is then held fixed during the test year. The

"adverse contribution" is the contribution of the component with the lowest pretest mean. It reflects both how often that component enters the tail and how severe its tail outcomes are.

I then ask whether the adverse contribution helps distinguish stocks that suffer large losses in bad markets, after controlling for total ES. The monthly stock-level regressions include size, recent mean returns, historical volatility, momentum, industry, market beta, downside beta, idiosyncratic volatility, and the predicted mean and volatility. A second regression compares the monthly contribution coefficients between bad and other market months. It includes fixed effects for each annual estimation window. A bad month has a next-month market excess return in the bottom quintile of the evaluation period. This rule selects 44 months, with a cutoff of -2.55%. The market classification is based on realized returns. The stock signals and portfolio positions are formed before those returns are observed.

With all controls included, a lower contribution is associated with a larger increase in crash risk during bad months in all three structured models. Sorting stocks by the contribution left after removing its linear relation with the controls gives a clear measure of the difference. In bad months, the lowest residual decile has a higher crash rate than the highest decile by 6.45, 7.14, and 6.59 percentage points at 5% ES for Structured K3, FiLM Sigma, and FiLM All. At 10% ES, the gaps are 6.46, 7.60, and 6.21 points. The corresponding other-month gaps are small and statistically insignificant at both levels in the main run. Two additional random seeds preserve the direction and concentration of this pattern in bad markets. Thus the decomposition helps distinguish stocks whose losses are more closely related to adverse market outcomes.

The paper makes three contributions. First, it introduces a structured forecasting design that gives common macroeconomic information and firm characteristics distinct roles. All three implementations improve density and tail forecasts relative to the fused neural benchmarks. Second, it shows how to use an exact ES decomposition to describe the composition of predicted tail losses. Third, it establishes that this composition contains economically useful information: the low-mean component contribution distinguishes stocks with different losses in bad markets after controlling for total ES, historical exposures, and predicted mean and volatility. The model therefore helps answer both how large a stock's predicted tail loss is and how its composition relates to losses during market declines.

1.1 Related Literature

This paper connects research on return-distribution forecasting, mixture models, and downside risk. Its main contribution is a structured MDN that assigns distinct roles to macroeconomic and firm information. The empirical analysis then uses the resulting forecasts to study both the size and the composition of stock-level tail risk.

1.1.1 Machine Learning and Return-Distribution Forecasts

Gu et al. (2020) show that machine learning can improve expected-return forecasts by capturing nonlinear predictor interactions. Distribution forecasts extend this task to the range of possible returns and their probabilities. Closely related to this paper, Barunik et al. (2026) use quantile neural networks and spline interpolation to forecast stock return distributions from firm and market predictors. Their approach illustrates how distribution forecasts can capture asymmetry and provide forecasts of moments as well as quantiles.

I study how an explicit division between common and firm-level information improves these forecasts. Macroeconomic inputs determine a single set of monthly component probabilities, while firm characteristics determine each stock's component means and standard deviations. This design produces a smooth density and an exact ES decomposition. The forecast comparisons evaluate its accuracy, and the comparison with a moment-matched Normal forecast tests the predictive information in distribution shape beyond the mean and variance.

1.1.2 Mixture Models and Structured Neural Networks

The method builds on Bishop (1994), who combines neural networks with mixture densities to model conditional distributions. The related mixture-of-experts literature makes both mixing weights and component predictions depend on inputs (Jordan and Jacobs, 1994). In financial econometrics, Giacomini et al. (2008) study flexible mixtures of t-distributions and demonstrate their usefulness in density forecasting and option pricing. These approaches establish the value of combining component distributions to represent a richer range of outcomes.

The structured MDN developed here gives the mixture a common monthly state across firms. Its probabilities depend on macroeconomic inputs, while returns within each state depend on firm information. The joint training objective evaluates the monthly cross-section under that common state. This links the organization of the inputs to the way the model learns from the stock panel. The two extensions use the feature-wise affine transformations of Perez et al. (2018) to let macroeconomic conditions adjust firm features.

Both retain the shared probabilities. Together, the three specifications examine the same structural idea under different treatments of macro–firm interactions.

1.1.3 Tail Risk and Stock Losses in Bad Markets

Kelly and Jiang (2014) use firm-level crashes to estimate common variation in tail risk and study its relation to asset prices. Ang et al. (2006) examine the relation between expected returns and sensitivity to downside market movements. These studies motivate a common description of market conditions together with measures of how stocks differ in their exposure to adverse outcomes. My model implements this division within a predictive density: state probabilities are shared, and the return distribution within each state varies across firms.

Firm-level tail-risk measures also contain information about subsequent returns. Huang et al. (2012) document a positive expected-return premium for their extreme downside-risk measure, while Atilgan et al. (2020) study left-tail momentum and its connection to underreaction and costly arbitrage. My analysis starts with forecasts of next-month return distributions and examines the stock losses that follow. Total-ES portfolios show whether the rankings distinguish downside performance. The contribution regressions and residual groups then ask whether the composition of predicted tail losses adds information after accounting for total ES, historical market and downside exposures, and predicted mean and volatility. This connects a forecast's component structure to realized vulnerability during market declines.

1.1.4 Expected Shortfall and Forecast Evaluation

The coherent-risk literature provides the foundation for using ES to summarize losses in the tail (Artzner et al., 1999; Acerbi and Tasche, 2002). I use the mixture representation to separate that summary into additive component contributions. Each contribution combines the component's share of predicted lower-tail outcomes with its mean return in that tail. The empirical question is whether these differences in composition help distinguish subsequent stock losses, alongside the information in total ES.

Proper scoring rules assess the full distribution (Gneiting and Raftery, 2007), weighted scores focus on selected tail regions (Gneiting and Ranjan, 2011), and PIT diagnostics assess probability calibration (Diebold et al., 1998). The joint elicitability results of Fissler and Ziegel (2016) and the VaR–ES models of Patton et al. (2019) provide the basis for joint tail-score comparisons. These tools evaluate the forecasts used in the stock-loss tests.

2 Data and Methods

2.1 Data

The estimation sample covers formation months from January 2000 through December 2024. Monthly returns, prices, market capitalization, and trading volume come from the Center for Research in Security Prices (CRSP). Accounting ratios and industry information come from Compustat through Wharton Research Data Services (WRDS). An observation must have a firm identifier, a valid formation-month return, positive capitalization, and a price of at least \$5. I exclude financial firms with Standard Industrial Classification (SIC) codes 6000–6999. The models use 21 continuous firm predictors: 15 accounting ratios, five derived market variables, and the raw stock return in the formation month. Industry enters through a six-dimensional learned embedding based on the Fama–French 12-industry classification. Missing industry classifications have a separate category.

Table 1: Firm-level information used in the density models

Group	Predictors
Valuation	Book-to-market, enterprise-value multiple, price-to-sales, price-to-cash-flow
Profitability and efficiency	Operating margin, cash-flow margin, return on equity, gross profitability, asset turnover
Leverage and liquidity	Debt-to-assets, short-term debt share, cash ratio, current ratio
Investment and accounting	R&D-to-sales, accruals
Market characteristics	Current monthly return, skip-month momentum, historical mean, historical volatility, turnover, log capitalization

Notes: The first four groups contain 15 accounting ratios. The last contains five derived market variables and the raw formation-month return, giving 21 continuous firm inputs. Industry enters separately through six learned features. Price is used for construction and eligibility, not as a separate neural input. Appendix A lists the fields.

I calculate historical variables on a complete calendar-month grid for each stock before applying the sample filters. A missing month therefore remains missing instead of being treated as the next observed record. The target is the raw CRSP return in the next calendar month. I check that stock–month records are unique and that the evaluation returns match the corresponding CRSP calendar months. The current return is observed at formation and enters directly as a predictor; it is distinct from this next-month target. Momentum compounds returns over the preceding 11 months, excluding the formation month, and requires at least six valid observations. Historical mean and volatility use

the 12 months ending at formation and also require at least six observations. Turnover is monthly volume divided by shares outstanding in consistent units. Size is log market capitalization.

Missing accounting ratios are filled first with the median for the same industry and formation month. If that median is unavailable, I use the median for all eligible firms in that month, then zero if needed. The ratios are winsorized: values below the monthly 1st percentile are set to that cutoff, and values above the 99th percentile are set to the upper cutoff. Missing derived variables use formation-month market medians; any remaining missing inputs use training-window medians. All continuous inputs are scaled with parameters estimated from the training sample. Appendix A lists the variables and explains their construction.

The macroeconomic inputs are 13 variables covering real activity, inflation, interest rates, credit markets, and equity markets. The monthly macroeconomic database FRED-MD, based on Federal Reserve Economic Data (FRED), supplies monthly vintage files; its variables use the specified transformations (McCracken and Ng, 2016). Each forecast month uses the latest available transformed observation in that vintage. Additional financial series come from FRED, and the S&P 500 return comes from WRDS. Accounting data are matched using their recorded public dates. For portfolio performance evaluation, I use the Fama–French five factors and the momentum factor to estimate risk-adjusted returns.

Table 2: Forecast and empirical evaluation samples

Sample	Stock–months	Months
Common seven-model forecast sample	480,446	216
Common eligible ES portfolio sample	480,429	216
History-complete common regression sample	410,049	216
Eligible rows excluded by historical requirements	70,380	—

Notes: Dates refer to formation months. Returns are observed from February 2007 through January 2025. Portfolio tests add date-matched CRSP ordinary-share and exchange requirements. Regressions also require valid historical market beta, downside beta, residual volatility, and all controls. Missing betas are not replaced with zero.

All seven models are compared on the same 480,446 stock–months over 216 formation months. Portfolio tests also require date-matched CRSP ordinary shares listed on the New York Stock Exchange (NYSE), the American Stock Exchange (AMEX), or Nasdaq. This leaves 480,429 observations. The conditional regressions require valid historical exposures and a complete control set, leaving 410,049 stock–months. All three structured specifications and every control specification use this same regression sample. The historical requirements remove stocks with insufficient histories in individual months.

2.2 Mixture Density Forecasts

2.2.1 Normal Mixtures

I use i to index stocks, t for the formation month, and $t+1$ for the next calendar month. The realized raw stock return is $R_{i,t+1}$; returns enter the formulas as decimals, so -0.10 means -10%. The information available at formation is denoted by $\mathcal{F}_t$. All forecasts condition on this information, even when the conditioning argument is omitted.

A single conditional Normal distribution is determined by its mean and variance. A mixture combines K component distributions, indexed by $k = 1, \ldots, K$. Here $K = 1$ for a neural Normal model and $K = 3$ for the three-component models. For a possible return value r, let $f_{i,t}(r)$ be stock i's predictive probability density:

$$f_{i,t}(r) = \sum_{k=1}^{K} \pi_{i,k,t} \frac{1}{\sigma_{i,k,t}} \phi\left(\frac{r - \mu_{i,k,t}}{\sigma_{i,k,t}}\right), \qquad \pi_{i,k,t} \geq 0, \quad \sum_{k} \pi_{i,k,t} = 1. \tag{1}$$

The weight $\pi_{i,k,t}$ is component k's probability, $\mu_{i,k,t}$ is its conditional mean, and $\sigma_{i,k,t} > 0$ is its conditional standard deviation. Thus $\sigma^2_{i,k,t}$ is a variance. The function ϕ is the standard Normal density, and Φ denotes its cumulative distribution function. A sum over k includes all K components. Differences in means and scales allow the mixture to be asymmetric, have more than one peak, and place more probability on large losses over relevant return ranges. Each component remains Normal.

I write $F_{i,t}(r) = \Pr(R_{i,t+1} \leq r \mid \mathcal{F}_t)$ for the predictive cumulative distribution function (CDF): the probability of a return no greater than r. Its inverse quantile function is $Q_{i,t}(u) = F^{-1}_{i,t}(u)$, where u is a probability level. Here Pr denotes probability; inverse CDFs return the cutoff associated with a given probability.

Let $\widehat{m}_{i,t}$ and $\widehat{v}_{i,t}$ denote the predictive mean and standard deviation implied by the fitted mixture. The hats mark quantities calculated from the fitted forecast. They are

$$\widehat{m}_{i,t} = \sum_{k} \pi_{i,k,t} \mu_{i,k,t}, \tag{2}$$

$$\widehat{v}_{i,t} = \left[\sum_{k} \pi_{i,k,t} (\sigma^2_{i,k,t} + \mu^2_{i,k,t}) - \widehat{m}^2_{i,t}\right]^{1/2}. \tag{3}$$

The firm subscript on $\pi_{i,k,t}$ allows the probabilities to differ across firms.

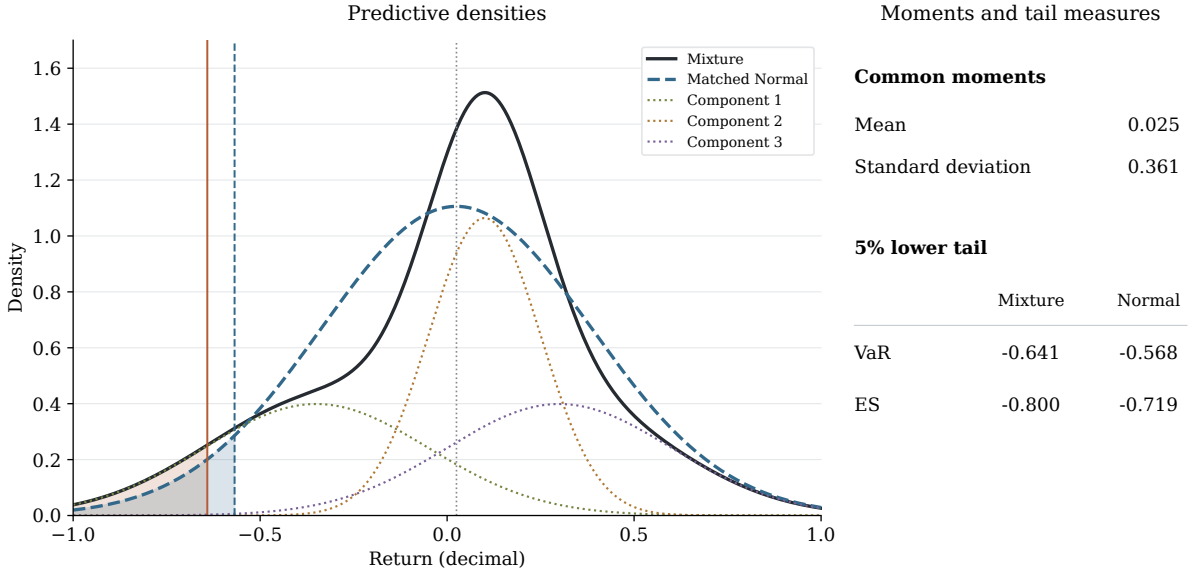
For a tail probability $0 < \alpha < 1$, define the lower-tail quantile, or VaR, by $q_{i,t,\alpha} = Q_{i,t}(\alpha)$. It is the return cutoff below which the forecast places probability α. Expected shortfall

averages the predicted return below that cutoff:

$$\mathrm{ES}_{i,t,\alpha} = \mathbb{E}_t[R_{i,t+1} \mid R_{i,t+1} \leq q_{i,t,\alpha}] = \frac{1}{\alpha}\int_0^{\alpha} Q_{i,t}(u)\, du. \tag{4}$$

The operator $\mathbb{E}_t[\cdot]$ denotes expectation conditional on $\mathcal{F}_t$; the additional condition in equation (4) restricts returns to the predicted lower tail. In the integral, u runs over probability levels from zero to α. A more negative ES means a more severe predicted tail loss. I report the main forecast and economic comparisons at both $\alpha = 0.05$ and $\alpha = 0.10$. The first describes a more extreme tail; the second averages over a wider part of the lower tail.

Figure 1: Tail Risk in a Normal Mixture and a Moment-Matched Normal



Notes: Colored dotted curves show the weighted Normal components; the solid black curve is their mixture. The blue dashed curve is a single Normal with the same mean and standard deviation. The orange solid and blue dashed vertical lines mark the respective 5% VaR cutoffs; the shaded regions show the lower tails. The gray dotted line marks the common mean. The right panel reports the common moments and the different tail measures. Returns are in decimal units. This is an illustrative example.

Figure 1 shows that the mean and variance do not fully describe tail risk. The mixture and the Normal substitute have the same mean and variance, but different lower-tail cutoffs and average tail returns. Section 3.5 uses the same idea to compare the fitted forecasts.

2.2.2 Structured MDN Parameterization

Let $x_{i,t} \in \mathbb{R}^{21}$ be the vector of 21 cleaned and scaled firm predictors, $e_{i,t} \in \mathbb{R}^6$ the six-dimensional learned industry embedding, and $m_t \in \mathbb{R}^{13}$ the vector of scaled macroeconomic predictors. The notation $\mathbb{R}^d$ means a vector of d real-valued entries. The firm network receives $x_{i,t}$ and $e_{i,t}$ concatenated into 27 entries (the $\oplus$ symbol in Figure 2); the industry embedding is not one of the 21 continuous predictors. Structured K3 uses these inputs as follows:

$$\pi_{k,t} = \pi_k(m_t), \qquad \mu_{i,k,t} = \mu_k(x_{i,t}, e_{i,t}), \qquad \sigma_{i,k,t} = \sigma_k(x_{i,t}, e_{i,t}). \tag{5}$$

The functions π_k, μ_k, and σ_k are the learned mappings from their indicated inputs to the corresponding component parameters. Dropping the firm index from $\pi_{k,t}$ indicates that the probabilities are shared. The resulting density is

$$f_{i,t}(r \mid \mathcal{F}_t) = \sum_{k=1}^{3} \pi_{k,t} \mathcal{N}(r; \mu_{i,k,t}, \sigma^2_{i,k,t}). \tag{6}$$

Here $\mathcal{N}(r; \mu, \sigma^2)$ is a Normal density evaluated at r, with mean μ and variance σ^2. A softmax transformation exponentiates the three probability-head scores and divides each by their sum. This produces nonnegative probabilities that add to one. The softplus function, $\text{softplus}(v) = \log(1 + \exp v)$ for a real input v, produces a positive value; adding 0.005 sets the scale floor. Throughout, log is the natural logarithm and exp its inverse. All firms share $\pi_{k,t}$ within month t, but their component means and standard deviations can differ. The structured model gives these inputs different roles. It also recognizes that observing more firms in a month does not provide more independent observations of that month's macroeconomic conditions.

The model assumes a common unobserved state $S_{t+1} \in \{1, 2, 3\}$. The state index identifies the component governing the cross-section of returns in month $t + 1$. Given $S_{t+1} = k$ and $\mathcal{F}_t$, each firm's return follows its own component Normal distribution, and returns are independent across firms. The joint return density is

$$f_t(\boldsymbol{r}) = \sum_{k=1}^{3} \pi_{k,t} \prod_{i \in \mathcal{U}_t} \mathcal{N}(r_i; \mu_{i,k,t}, \sigma^2_{i,k,t}), \tag{7}$$

Here $\boldsymbol{r} = (r_i)_{i \in \mathcal{U}_t}$ is a vector of possible returns, $\mathcal{U}_t$ is the set of firms included at formation, and $N_t = |\mathcal{U}_t|$ is its size. Vertical bars around a set denote its number of elements; $\prod$ means multiplication over the indicated firms. I write $\mathbf{R}_{t+1} = (R_{i,t+1})_{i \in \mathcal{U}_t}$ for the corresponding

realized return vector. Each stock's individual distribution is given by equation (6). In equation (7), all firms share the same latent state, which links their returns through the common monthly environment.

The two FiLM extensions retain the shared probabilities and allow macroeconomic conditions to adjust firm features before the model calculates component parameters. They provide two additional implementations of the common structure. Let $h_{i,t} = h(x_{i,t}, e_{i,t}) \in \mathbb{R}^8$ be the firm's eight learned features and $g_t = g(m_t) \in \mathbb{R}^8$ the eight macro features. The functions h and g are the learned firm and macro encoders. As shown in Figure 2, the FiLM layer uses g_t to produce two eight-dimensional vectors: a *scale change* $\Delta\gamma_t$ and a *shift* β_t:

$$\Delta\gamma_t = 0.1 \tanh\{a(g_t)\}, \qquad \beta_t = 0.1 \tanh\{b(g_t)\}. \tag{8}$$

The functions a and b are learned linear transformations with intercepts, and the hyperbolic tangent tanh, which maps real inputs to values between -1 and 1, is applied to each element. The scale change $\Delta\gamma_t$ modifies the multiplier on each firm feature; the shift β_t adds to that feature. Together they form the adjusted features:

$$\widetilde{h}_{i,t} = (1 + \Delta\gamma_t) \odot h_{i,t} + \beta_t, \tag{9}$$

$$\sigma_{i,k,t} = \text{softplus}\{s_k(\widetilde{h}_{i,t})\} + 0.005. \tag{10}$$

The operator $\odot$ means element-by-element multiplication, and $1 + \Delta\gamma_t$ adds one to each entry of the scale-change vector. The functions s_k and u_k are the learned scale and mean output heads, respectively. Both $\Delta\gamma_t$ and β_t are shared by all firms in month t. The same adjustment can affect firms differently because their original features $h_{i,t}$ differ. The FiLM coefficient layer is initialized at zero, giving $\Delta\gamma_t = \beta_t = 0$ and hence $\widetilde{h}_{i,t} = h_{i,t}$ at initialization. FiLM Sigma uses the original features for means, $\mu_{i,k,t} = u_k(h_{i,t})$. FiLM All uses the adjusted features, $\mu_{i,k,t} = u_k(\widetilde{h}_{i,t})$. Both models keep the macro probability head and common-state assumption of Structured K3. Bounding the scale change and shift allows macro–firm interactions without removing the baseline structure.

Figure 2 shows the common architecture and the FiLM extensions. The macro representation g_t feeds the probability head in every specification. In the FiLM models, a second branch produces the scale change $\Delta\gamma_t$ and shift β_t that transform $h_{i,t}$ into $\widetilde{h}_{i,t}$. The original firm representation remains unchanged. Both feature vectors enter the input table, which specifies their use in each model: Structured K3 uses $h_{i,t}$ for both heads, FiLM Sigma uses $\widetilde{h}_{i,t}$ only for scale, and FiLM All uses it for both heads.

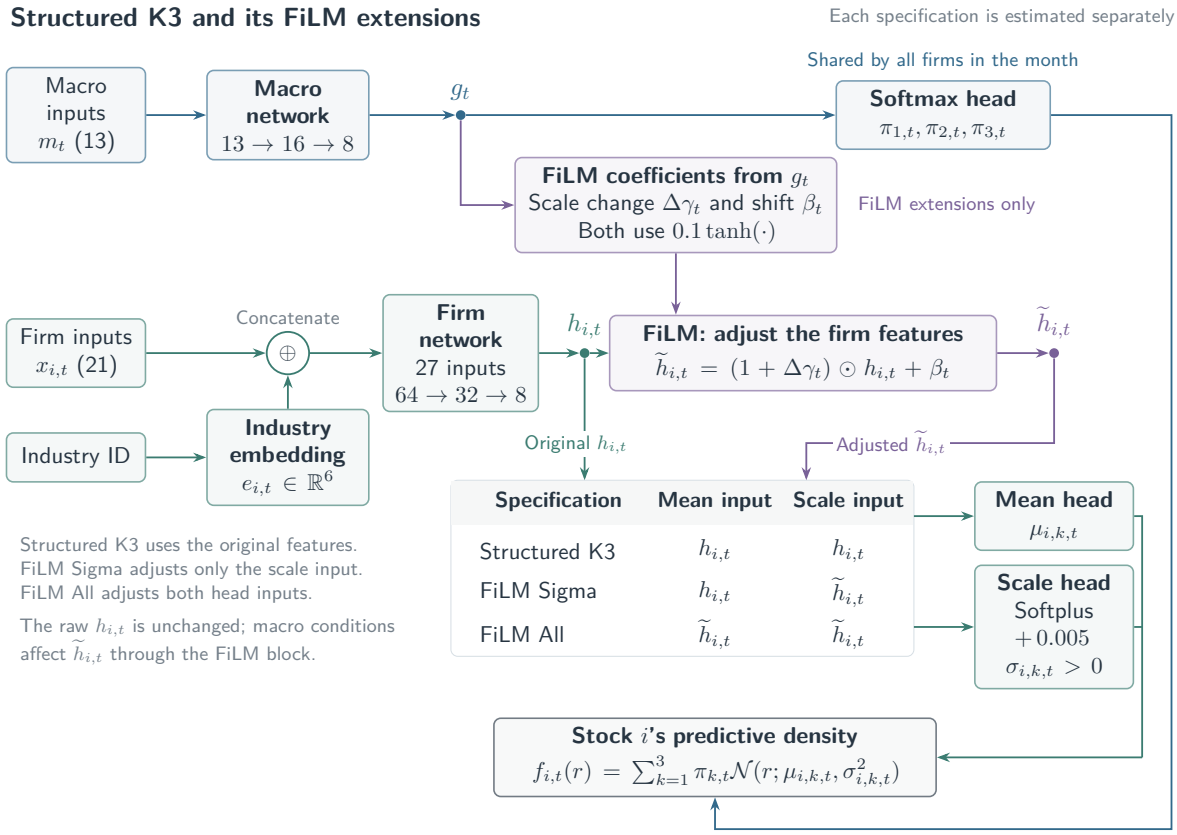


Figure 2: Structured K3 and macro-conditioned FiLM extensions

Notes: Macro features g_t determine the common probabilities in every specification. In the FiLM extensions, they also generate the scale change $\Delta\gamma_t$ and shift β_t used to form $\widetilde{h}_{i,t}$ from the original firm features $h_{i,t}$. The input table specifies which features enter each head. The FiLM path is absent in Structured K3. All specifications retain three Normal components and are estimated separately. Feature multiplication is element by element; the same monthly adjustment vectors are applied to every firm.

2.2.3 Rolling Estimation and Training

Figure 3 shows the annual estimation schedule. Each window has 72 training months, followed by 12 validation months and 12 test months. In the first window, formation years 2000–2005 are used for training, 2006 for validation, and 2007 for testing. All three blocks then move forward by 12 months. The last window trains on 2017–2022, validates on 2023, and tests on 2024. This gives 18 test windows, indexed by $w = 0, \ldots, 17$ in the formulas; the figure numbers these same windows from 1 to 18. Let $\mathcal{T}_w$ be the 72 training months and $\mathcal{A}_w$ the 84 training and validation months for window w. A date index such as s within these sets denotes a formation month; N_s is the number of firms in that month. In Figure 3, $\bar{\mu}_{k,w}$ is component k's fitted mean averaged first across firms and then across $\mathcal{A}_w$. The plain symbol A_w is the index of the component with the lowest such mean; arg min selects that minimizing index. Thus A_w is a component label, while the calligraphic $\mathcal{A}_w$ is a set of months. The dates refer to formation months: inputs in month t predict the stock return in month $t + 1$.

Panel B separates model fitting, model selection, component labeling, and forecasting. Every neural model starts from a new initialization in each window. Early stopping selects the saved model with the lowest validation loss. The main specification does not retrain this model on the combined training and validation data. For the structured mixtures, the selected model is then evaluated on the 84 training and validation months to rank its component means. The lowest-mean component is labeled adverse, as defined in Section 5.1. This step assigns labels; it does not update model parameters. Both the parameters and labels remain fixed during the next 12 test months, while forecasts change with each month's inputs. Labels are determined again after fitting the next window.

For continuous firm predictor $j = 1, \ldots, 21$, let $x^{\text{imp}}_{i,j,t}$ be the value after the predictor-specific cleaning and missing-value treatment described in Section 2.1, but before scaling. The label imp means "imputed" and includes any applicable prior winsorization; it is not an exponent. Let $\overline{x}_j^{\text{tr},w}$ and $s_j^{\text{tr},w}$ be its pooled training-window mean and sample standard deviation. The label tr means "training," and w identifies the window. The scaled value, with sc meaning "scaled," is

$$x^{\text{sc}}_{i,j,t} = \frac{x^{\text{imp}}_{i,j,t} - \overline{x}_j^{\text{tr},w}}{s_j^{\text{tr},w} + 10^{-8}}. \tag{11}$$

The subscripts i, j, and t identify the stock, predictor, and formation month. The mean and sample standard deviation are calculated across all training stock–months in window w, rather than separately within each month. Subtracting this mean and dividing by this standard deviation puts predictors on a common scale. The small constant 10^{-8} prevents

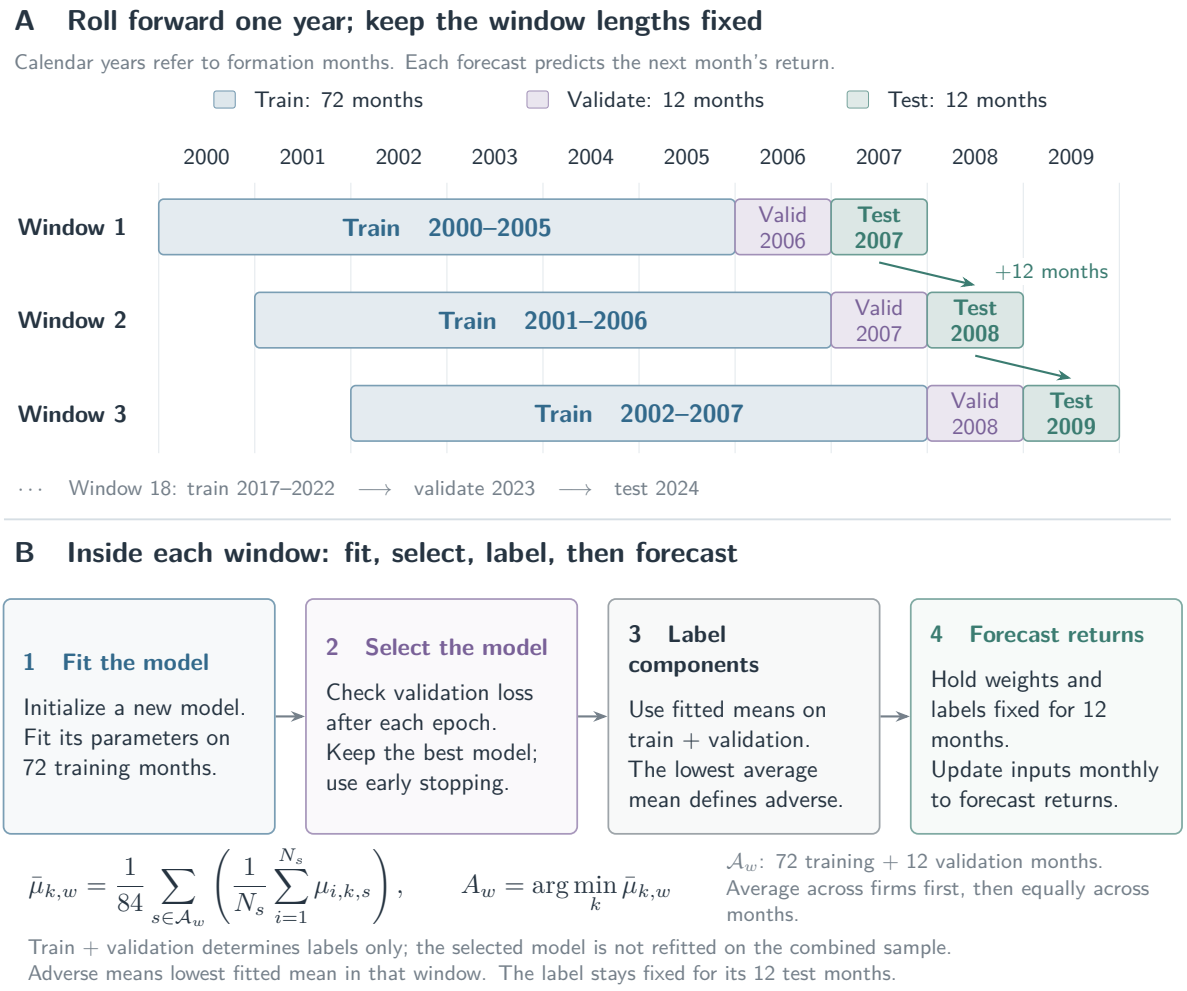


Figure 3: Annual rolling estimation and component labeling. Each window uses six years for training, one for validation, and one for testing. All blocks advance by one year, producing 18 test windows over formation years 2007–2024. Each model starts from a new initialization. Validation selects the best checkpoint, with no subsequent refit. Using that checkpoint, component means are averaged first across firms and then equally across the 84 training and validation months ($\mathcal{A}_w$). The lowest-mean component is labeled adverse. Parameters and labels are fixed during the test year; forecasts vary with monthly inputs. Labels are recomputed in each new window. Formation month t predicts the return in $t+1$.

division by zero when a training predictor has no variation. The neural input $x_{i,t}$ contains these scaled entries. Validation and test observations use the same training-window quantities, without re-estimating them on later data. Macro variables are scaled using one observation per training month, and missing values use training medians. Return targets remain in their original units. These forecasting transformations are separate from the monthly transformations used in the later regressions.

The structured training objective has two parts: a normalized joint negative log likelihood (NLL) and a marginal NLL. Figure 4 shows how both are calculated from the same model outputs. Write $f_{i,k,t}(r) = \mathcal{N}(r; \mu_{i,k,t}, \sigma^2_{i,k,t})$ for firm i's component-k density, before weighting by the common probability $\pi_{k,t}$. Let $\overline{N}_w = |\mathcal{T}_w|^{-1} \sum_{t \in \mathcal{T}_w} N_t$ be the average number of firms per training month in window w, where $|\mathcal{T}_w|$ counts the training months.

The left-hand path in the figure evaluates the joint return distribution. Within each state, it multiplies the component densities of all firms. It then weights these products by the common state probabilities and sums across states. The right-hand path evaluates each stock's marginal distribution: it first mixes the three component densities for each firm and then averages the individual log losses across firms. The two monthly losses are

$$\ell_t^J = -\frac{1}{\overline{N}_w} \log \left[\sum_{k=1}^{3} \pi_{k,t} \prod_{i \in \mathcal{U}_t} f_{i,k,t}(R_{i,t+1}) \right], \tag{12}$$

$$\ell_t^M = -\frac{1}{N_t} \sum_{i \in \mathcal{U}_t} \log \left[\sum_{k=1}^{3} \pi_{k,t} f_{i,k,t}(R_{i,t+1}) \right]. \tag{13}$$

The superscripts J and M label the joint and marginal terms; they are not powers. "Marginal" means that each stock's distribution is evaluated separately, while the joint term evaluates the return vector under the common-state model. The joint NLL is divided by the average training-month firm count to put it on a comparable scale to the average individual-stock NLL. Both paths use every firm and all three states. Neither assigns a firm to a single component.

I average each loss equally across training months. The window-level marginal loss is

$$\mathcal{L}_w^M = \frac{1}{|\mathcal{T}_w|} \sum_{t \in \mathcal{T}_w} \ell_t^M = -\frac{1}{|\mathcal{T}_w|} \sum_{t \in \mathcal{T}_w} \frac{1}{N_t} \sum_{i \in \mathcal{U}_t} \log f_{i,t}(R_{i,t+1}). \tag{14}$$

The joint loss and combined training objective are

$$\mathcal{L}_w^J = \frac{1}{|\mathcal{T}_w|} \sum_{t \in \mathcal{T}_w} \ell_t^J, \tag{15}$$

$$\mathcal{L}_w^S = 0.9\mathcal{L}_w^J + 0.1\mathcal{L}_w^M = \frac{1}{|\mathcal{T}_w|} \sum_{t \in \mathcal{T}_w} \left(0.9\ell_t^J + 0.1\ell_t^M\right). \tag{16}$$

Lowercase ℓ denotes a monthly loss, whereas uppercase $\mathcal{L}$ denotes its average across a window. The superscript S labels the combined structured objective. As shown at the bottom of the figure, the objective places weight 0.9 on the normalized joint term and 0.1 on the marginal term. The first evaluates the common-state joint distribution; the second directly evaluates individual return distributions. Validation uses the same combination and the training value of $\overline{N}_w$. Each training batch contains complete monthly cross-sections; no firms are subsampled. Appendix B gives the network and optimization settings.

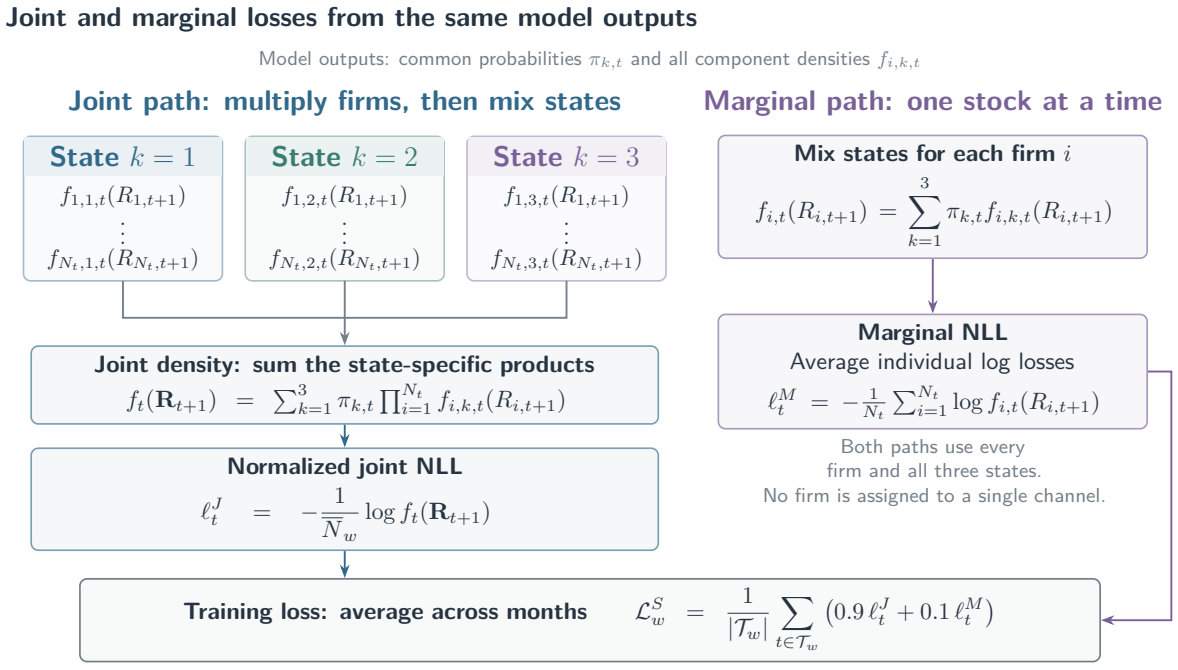


Figure 4: Common-state joint NLL and individual marginal NLL

Notes: The joint path multiplies all firms' component densities within each state and then mixes across states. The marginal path first mixes states for each firm and then averages individual negative log likelihoods. Here N_t is the monthly firm count and $\overline{N}_w$ is its training-window average, held fixed for validation. The structured objective combines the normalized joint and marginal terms with weights 0.9 and 0.1 and averages equally across months. Both paths retain every firm and all three states.

3 Model Comparison and Forecast Evaluation

I compare seven models: Rolling Normal, Linear Quantile, Fused K1, Fused K3, Structured K3, FiLM Sigma, and FiLM All. Rolling Normal uses each stock's past 84 calendar months of returns. Linear Quantile estimates return quantiles from a linear function of the common predictors. Fused K1 uses a neural network but keeps a single Normal distribution. Fused K3 adds three mixture components without separating the roles of macro and firm inputs. The structured models make this separation, and FiLM allows macro conditions to adjust the firm features. Appendix B describes the benchmarks.

Table 3: Model designs and the role of each input set

Model	Probabilities	Component means	Component scales
Rolling Normal	Single component	Own return history	Own return history
Linear Quantile	Not applicable	Direct quantile forecasts	Direct quantile forecasts
Fused K1	Single component	Fused macro and firm	Fused macro and firm
Fused K3	Fused macro and firm	Fused macro and firm	Fused macro and firm
Structured K3	Common macro probabilities	Firm features	Firm features
FiLM Sigma	Common macro probabilities	Firm features	Macro-adjusted firm features
FiLM All	Common macro probabilities	Macro-adjusted firm features	Macro-adjusted firm features

Notes: All neural density models use the same 21 firm variables, 13 macro variables, and six-dimensional industry embedding. Fused K1 has one Normal component; the other neural densities have three. The structured models use one common monthly state and the training loss in equation (16). Linear Quantile fits a linear predictor for each grid quantile.

3.1 Forecast Quantiles and Numerical Checks

The models are evaluated on the common quantile grid $\mathcal{G}$, a set of 27 probability levels rather than return values:

$$\mathcal{G} = \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.10, 0.15, 0.25, 0.40, 0.50,$$
$$0.60, 0.75, 0.85, 0.90, 0.91, 0.92, 0.93, 0.94, 0.95, 0.96, 0.97, 0.98, 0.99\}.$$

For a Normal distribution, quantiles use the inverse standard Normal CDF Φ^{-1}, which returns the cutoff associated with a given probability. Mixture quantiles solve

$$F_{i,t}(q) = \sum_k \pi_{i,k,t} \Phi\left(\frac{q - \mu_{i,k,t}}{\sigma_{i,k,t}}\right) = \alpha \tag{17}$$

Here the unknown q is $q_{i,t,\alpha}$, and $F_{i,t}$ is the mixture CDF obtained by summing the weighted component CDFs. Bisection finds the cutoff by repeatedly halving an interval that brackets the desired probability. For Linear Quantile, I sort the predicted quantiles within each stock–month to prevent crossings. For grid-based scores, the smallest and largest predicted quantiles are held constant out to probabilities zero and one. The same approximation is used for every model, including those without an exact density. In the main comparison, stored VaR predictions agree with independently calculated quantiles to within 5×10^{-12}.

3.2 Forecast Accuracy Metrics

Let $\ell_{i,t}$ be a forecast loss for stock i formed at t and evaluated using the observed return in $t+1$. With T evaluation months, the reported average is $T^{-1} \sum_t N_t^{-1} \sum_i \ell_{i,t}$. Here $T = 216$, and N_t is the number of stocks in the relevant common sample in month t; any score-specific validity restriction is stated below. I first average across stocks within each month, then average across months. Statistical tests use the monthly series, so precision is based on the number of months rather than treating all stock–months as independent.

3.2.1 Negative Log Likelihood

The log score is

$$\mathrm{NLL}_{i,t} = -\log f_{i,t}(R_{i,t+1}). \tag{18}$$

A lower NLL means that the model assigns more density near the observed return. A continuous density can exceed one, so NLL values can be negative. NLL can be calculated for each density model. It is not reported for Linear Quantile because a finite grid of quantiles does not define a unique smooth density.

3.2.2 Quantile Score

Let $u_{i,t,\alpha} = R_{i,t+1} - q_{i,t,\alpha}$ be the quantile forecast error. Write $\mathbf{1}\{A\}$ for an indicator equal to one when condition A holds and zero otherwise. Define the pinball loss $\rho_\alpha(u) =$

$u[\alpha - \mathbf{1}\{u < 0\}]$. The quantile score is

$$\mathrm{QS}_{i,t,\alpha} = 2\rho_\alpha(u_{i,t,\alpha}). \tag{19}$$

The abbreviation QS means quantile score; the factor of two is its reporting convention. On average, this score is lowest when the forecast is the correct conditional quantile (Koenker and Bassett, 1978). I report results at the 5% and 10% levels.

3.2.3 Continuous Ranked Probability Score

CRPS evaluates the full predicted CDF. In the following expression, F is a generic predictive CDF, y the observed return, z a possible return used as an integration variable, and $Q = F^{-1}$ the corresponding quantile function. In the second integral, u is a probability level and ρ_u is the pinball loss at that level:

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} [F(z) - \mathbf{1}\{y \le z\}]^2 \, dz \tag{20}$$

$$= 2 \int_0^1 \rho_u\{y - Q(u)\} \, du. \tag{21}$$

For the seven-model comparison, I approximate this integral with the trapezoidal rule on the common quantile grid, including the endpoint extensions. In the matched-Normal exercise in Section 3.5, both forecasts have exact densities, so I calculate CRPS analytically.

3.2.4 Lower-Tail CRPS

The normalized lower-tail score is

$$\mathrm{TailCRPS}_{i,t}^{0:\alpha} = \frac{2}{\alpha} \int_0^\alpha \rho_u\{R_{i,t+1} - Q_{i,t}(u)\} \, du. \tag{22}$$

The superscript $0:\alpha$ denotes the probability interval from zero to α, rather than a power. Every model uses the same grid and integration rule. Dividing by α gives an average score over the selected tail range. Lower values mean better lower-tail forecasts (Gneiting and Ranjan, 2011).

3.2.5 Joint VaR–ES Score

For one stock and formation month, abbreviate the observed return as $y = R_{i,t+1}$, the VaR forecast as $q = q_{i,t,\alpha}$, and the ES forecast as $e = \mathrm{ES}_{i,t,\alpha}$. The joint scoring function,

denoted by L_{FZ0}, is

$$L_{FZ0}(y, q, e; \alpha) = -\frac{\mathbf{1}\{y \leq q\}(q - y)}{\alpha e} + \frac{q}{e} + \log(-e) - 1. \tag{23}$$

ES enters in its original signed return units. The score is calculated only when $q < 0$, $e < 0$, and $e \leq q$. At each tail level, I retain only observations for which all seven models satisfy these conditions. This keeps 98.860% of the common sample at 5% and 97.344% at 10%. Table 4 reports the valid share for each model. A lower FZ0 score means that VaR and ES are more accurate as a pair; it does not evaluate ES alone (Fissler and Ziegel, 2016; Patton et al., 2019).

Table 4: Validity of joint VaR–ES scores

Model	Valid 5% (%)	Valid 10% (%)
Rolling Normal	99.867	99.806
Linear Quantile	98.992	97.664
Fused K1	100.000	99.909
Fused K3	100.000	99.922
Structured K3	100.000	100.000
FiLM Sigma	100.000	100.000
FiLM All	100.000	100.000

Notes: A forecast is valid for FZ0 when $q < 0$, $e < 0$, and $e \leq q$. Keeping only observations valid for all seven models retains 98.860% of stock–months at 5% and 97.344% at 10%.

3.2.6 Forecast Tables

Table 5 shows a common improvement across the three structured models. Their NLL values range from −0.6608 to −0.6406, compared with −0.5029 for Fused K1 and −0.5051 for Fused K3. Their CRPS values are 0.07123–0.07222, compared with 0.07803 and 0.08132. Thus the CRPS reductions are approximately 7.4–8.7% relative to Fused K1 and 11.2–12.4% relative to Fused K3.

The gains also extend to the lower tail. All three structured models have lower 5% and 10% tail CRPS, quantile scores, and joint VaR–ES scores than both fused benchmarks. Their 5% FZ0 values lie between −1.3701 and −1.3516, compared with −1.2173 for Fused K1 and −1.1774 for Fused K3. At 10%, their values are −1.5653 to −1.5489, compared with −1.4077 and −1.3606. Sharing the monthly probabilities is common to all three designs; allowing macroeconomic inputs to adjust firm features provides alternative ways to retain that structure. These forecast gains hold across all three structured variants.

Table 5: Out-of-sample distribution and tail forecast losses

Model	NLL	CRPS	Tail 5%	Tail 10%	QS 5%	QS 10%	FZ0 5%	FZ0 10%
Rolling Normal	61486.5495$^{(6)}$	0.0735$^{(4)}$	0.0179$^{(6)}$	0.0286$^{(5)}$	0.0301$^{(5)}$	0.0478$^{(4)}$	-1.2190$^{(4)}$	-1.4818$^{(4)}$
Linear Quantile	—	0.0863$^{(7)}$	0.0211$^{(7)}$	0.0361$^{(7)}$	0.0375$^{(7)}$	0.0623$^{(7)}$	-1.1736$^{(7)}$	-1.3987$^{(6)}$
Fused K1	-0.5029$^{(5)}$	0.0780$^{(5)}$	0.0170$^{(4)}$	0.0284$^{(4)}$	0.0297$^{(4)}$	0.0490$^{(5)}$	-1.2173$^{(5)}$	-1.4077$^{(5)}$
Fused K3	-0.5051$^{(4)}$	0.0813$^{(6)}$	0.0176$^{(5)}$	0.0293$^{(6)}$	0.0308$^{(6)}$	0.0503$^{(6)}$	-1.1774$^{(6)}$	-1.3606$^{(7)}$
Structured K3	-0.6406$^{(3)}$	0.0722$^{(3)}$	0.0159$^{(2)}$	0.0265$^{(2)}$	0.0278$^{(2)}$	0.0457$^{(3)}$	-1.3648$^{(2)}$	-1.5633$^{(2)}$
FiLM Sigma	-0.6608$^{(1)}$	0.0712$^{(1)}$	0.0158$^{(1)}$	0.0263$^{(1)}$	0.0277$^{(1)}$	0.0453$^{(1)}$	-1.3701$^{(1)}$	-1.5653$^{(1)}$
FiLM All	-0.6475$^{(2)}$	0.0717$^{(2)}$	0.0159$^{(3)}$	0.0265$^{(3)}$	0.0279$^{(3)}$	0.0456$^{(2)}$	-1.3516$^{(3)}$	-1.5489$^{(3)}$

Notes: Formation dates run from January 2007 through December 2024: 216 months and 480,446 common stock–months. Lower scores mean better forecasts. CRPS and tail CRPS use the same quantile grid for all models. FZ0 uses observations where all models have valid negative-ES forecasts. Linear Quantile has no exact density, so NLL is not reported. Rolling Normal NLL is large because it assigns very little density to some observed returns. Section 3 defines the scores. Superscripts in parentheses report ranks within each score column (1 is best), computed from unrounded losses among models with that score available. Exact ties share a rank; rounded values can be identical despite different ranks.

3.2.7 Tests of Forecast Differences

Let the superscripts a and b identify a candidate model and a benchmark, and let $\ell^a_{i,t}$ and $\ell^b_{i,t}$ be their losses for the same observation. Their average monthly loss difference is

$$d_t^{a,b} = \frac{1}{N_t} \sum_i \left[\ell^a_{i,t} - \ell^b_{i,t} \right]. \tag{24}$$

A negative average difference means that the candidate forecasts better. Here $\mathbb{E}$ denotes the time-series expectation over evaluation months. I test $\mathbb{E}[d_t^{a,b}] = 0$ with Newey–West standard errors using 12 lags (Newey and West, 1987). The test compares monthly loss differences, following Diebold and Mariano (1995). I abbreviate Newey–West inference with 12 calendar-month lags as NW12. These standard errors allow heteroskedasticity and serial correlation, hence the term heteroskedasticity-and-autocorrelation-consistent (HAC). In tables, SE means estimated standard error, and CI means confidence interval. The reported p is the two-sided asymptotic p-value for the stated null hypothesis.

Table 6 compares each structured model with each fused benchmark. Every reported NLL, CRPS, and FZ0 difference at both tail levels is negative and statistically significant at 5%. Figure 5 displays CRPS and both FZ0 comparisons with the unrestricted mixture. These results support the structured family across different component parameterizations. The gains are consistent across the three implementations of the structured forecasting design.

Table 6: Paired predictive-accuracy comparisons at both tail levels

Candidate	Benchmark	Δ NLL	p	Δ CRPS	p	Δ FZ0 5%	p	Δ FZ0 10%	p
Structured K3	Fused K1	-0.1377	< 0.001	-0.0058	0.005	-0.1474	< 0.001	-0.1556	< 0.001
Structured K3	Fused K3	-0.1355	0.006	-0.0091	0.004	-0.1873	< 0.001	-0.2027	< 0.001
FiLM Sigma	Fused K1	-0.1579	< 0.001	-0.0068	0.001	-0.1528	< 0.001	-0.1577	< 0.001
FiLM Sigma	Fused K3	-0.1557	0.004	-0.0101	0.003	-0.1927	< 0.001	-0.2047	< 0.001
FiLM All	Fused K1	-0.1446	< 0.001	-0.0064	0.002	-0.1343	< 0.001	-0.1412	< 0.001
FiLM All	Fused K3	-0.1423	0.008	-0.0097	0.004	-0.1742	< 0.001	-0.1883	< 0.001

Notes: Differences are candidate loss minus benchmark loss. Negative differences favor the candidate. NLL and CRPS evaluate the full distribution; FZ0 is reported separately at both tail levels. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

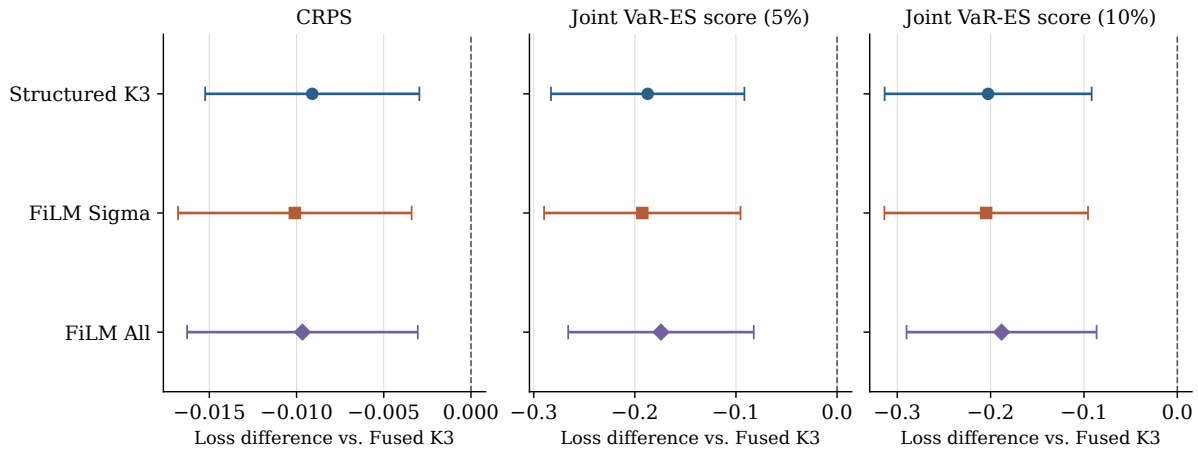


Figure 5: Predictive-accuracy gains relative to the unrestricted mixture

Notes: Points show each model's average loss minus Fused K3 loss over 216 formation months. Lines show pointwise 95% confidence intervals based on NW12 standard errors. Negative values favor the structured model. The two FZ0 panels use their respective common valid samples at the 5% and 10% tail levels.

3.3 Probability Integral Transform

For a continuous forecast, the probability integral transform is $U_{i,t} = F_{i,t}(R_{i,t+1})$. The symbol $U_{i,t}$ is the PIT value. It records where the observed return falls within the predicted distribution. If the conditional distribution is correct, PIT values are uniform between zero and one. Density models use their exact CDFs. Linear Quantile uses interpolation between its sorted quantiles. I calculate the PIT mean, variance, and Kolmogorov–Smirnov distance within each month, then average across months. I also calculate the first-order autocorrelation of the monthly PIT mean to measure its persistence. Table 7 abbreviates the Kolmogorov–Smirnov distance as KS and this one-month-lag autocorrelation as ACF(1). Coverage is the fraction of realized returns below the predicted VaR cutoff, averaged equally across months.

Table 7 shows three calibration improvements shared by all three structured models relative to both fused benchmarks. First, their average PIT values, 0.5091–0.5259, are closer to the target of 0.5 than the fused values of 0.5461 and 0.5490. Actual returns therefore occupy a more balanced position within the structured models' predicted distributions. Second, their average monthly KS distances fall to 0.2023–0.2074, compared with 0.2467 and 0.2572 for the fused models. On this measure, their PIT distributions are closer to the uniform reference. Third, the first-order autocorrelation of monthly average PIT falls from about 0.26 for the fused models to values close to zero, between -0.054 and -0.005. Months with high or low average PIT are therefore less likely to be followed by another month with the same deviation. These descriptive comparisons show that the structured models' gains extend beyond lower forecast losses: their predicted probabilities are better centered and show less persistent deviations in monthly average PIT.

The structured models also have average VaR exceedance rates close to their nominal targets: 4.748%–4.905% at the 5% level and 8.764%–9.035% at the 10% level. NW12 tests do not reject the respective targets for either the structured models or the benchmarks. Mean VaR coverage is therefore close to the targets across both model groups. The 10% exceedance rates remain below their target, and PIT variances remain below the uniform reference of 1/12. Across the three PIT diagnostics discussed above, the structured models improve on the fused benchmarks. Figure 6 displays the corresponding PIT histograms.

Table 7: PIT diagnostics and VaR coverage at both tail levels

Model	PIT mean	Variance	KS	ACF(1)	Cov. 5%	p	Cov. 10%	p
Rolling Normal	0.4827	0.0597	0.1985	-0.0511	5.393	0.640	9.409	0.591
Linear Quantile	0.5253	0.0660	0.2481	0.0458	6.353	0.294	11.508	0.407
Fused K1	0.5461	0.0585	0.2467	0.2622	5.032	0.969	8.737	0.299
Fused K3	0.5490	0.0662	0.2572	0.2558	6.679	0.109	11.524	0.332
Structured K3	0.5259	0.0621	0.2074	-0.0049	4.847	0.806	8.764	0.175
FiLM Sigma	0.5091	0.0605	0.2023	-0.0544	4.748	0.650	8.789	0.137
FiLM All	0.5091	0.0610	0.2043	-0.0272	4.905	0.869	9.035	0.258

Notes: Uniform PIT targets are mean 0.5 and variance 1/12. KS is the mean monthly cross-sectional distance from uniformity; ACF(1) uses monthly PIT means. Coverage entries are equal-month average exceedance percentages, tested against their respective 5% or 10% targets with NW12 standard errors. These are unconditional mean-coverage tests.

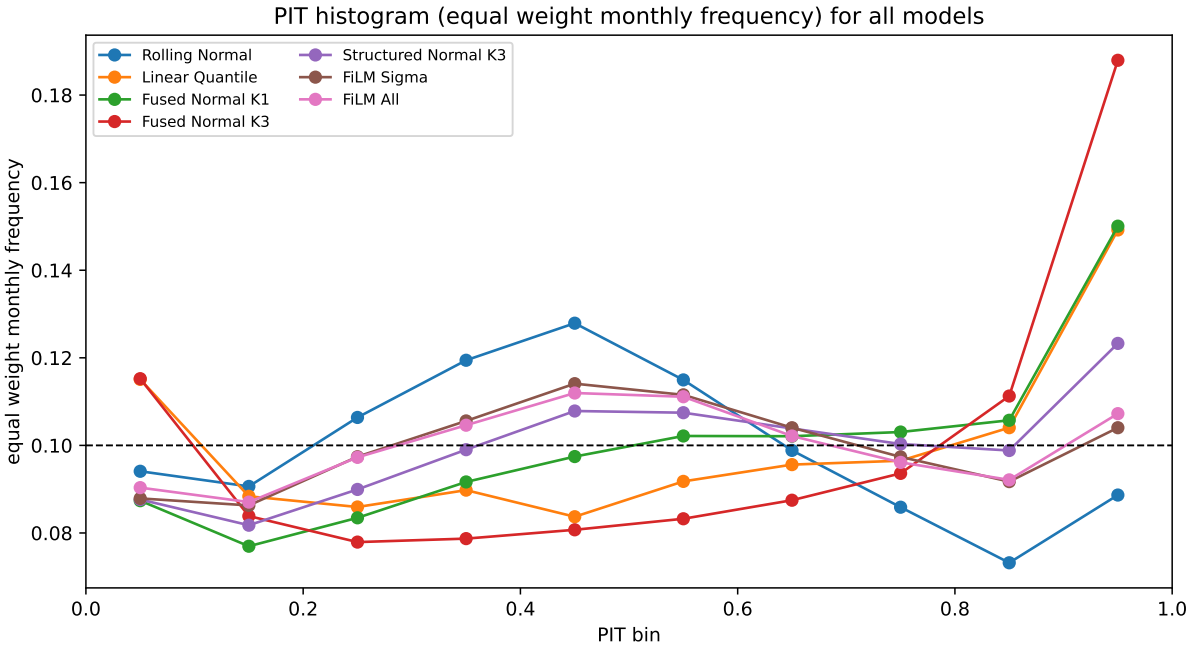


Figure 6: Probability integral transform histograms

Notes: Bin shares are calculated within each formation month, then averaged equally over the 216 months. A correctly calibrated continuous forecast has a uniform PIT, placing 10% of observations in each of the ten bins. Linear Quantile uses interpolation between its sorted quantiles.

3.4 Overlap in ES-Based Stock Selection

The three structured models can differ in their forecast parameters while selecting similar stocks at the extremes of predicted tail risk. To examine this directly, I form each model's lowest and highest ES deciles within the common seven-model forecast universe. The lowest decile contains stocks with the most negative ES; the highest contains stocks with the least severe predicted tail losses. These groups each contain approximately 10% of stocks. They are distinct from the 5% or 10% probability level used to calculate ES.

For models a and b, let $\mathcal{D}^a_{g,t}$ and $\mathcal{D}^b_{g,t}$ be their selected stock sets for extreme group g in month t. I measure their overlap by

$$O^{a,b}_{g,t} = \frac{2|\mathcal{D}^a_{g,t} \cap \mathcal{D}^b_{g,t}|}{|\mathcal{D}^a_{g,t}| + |\mathcal{D}^b_{g,t}|}. \tag{25}$$

Here g identifies the lowest or highest ES stock decile, $\cap$ is the intersection of two stock sets, and $O^{a,b}_{g,t}$ is their monthly Dice overlap. A separate comparison is made at each ES probability level, whose index is suppressed. This coefficient equals the fraction of selected stocks shared by the models when the two sets have the same size. It equals one for identical selections and zero for disjoint selections. I report its average over the 216 formation months, with equal weight for each month. Independent random selections of one decile would overlap by approximately 10% under this measure.

Figure 7 shows that the three structured models agree much more closely with one another than any other pair of models in this seven-model comparison. At 5% ES, their lowest-decile overlaps range from 88.2% to 90.7%; at 10% ES, they range from 88.0% to 90.6%. No other model pair exceeds 77.6% and 77.2%, respectively. The same separation appears in the highest stock decile: within-family overlaps are 91.2%–93.7% at 5% ES and 90.8%–93.4% at 10% ES, while the largest overlaps among all remaining model pairs are 76.8% and 74.1%. These remaining pairs include structured models paired with any of the four benchmarks, as well as pairs formed only from benchmarks. In each of the four panels, even the smallest overlap within the structured family exceeds the largest overlap outside that block, excluding the diagonal self-comparisons.

The economic meaning is direct: the three structured specifications select largely the same stocks at both extremes of predicted tail risk, despite their different mean and scale parameterizations. Switching to another model changes stock selection much more than switching between structured variants. This evidence supports treating the three structured specifications as a closely related family.

The closer within-family agreement relative to the fused benchmarks also appears

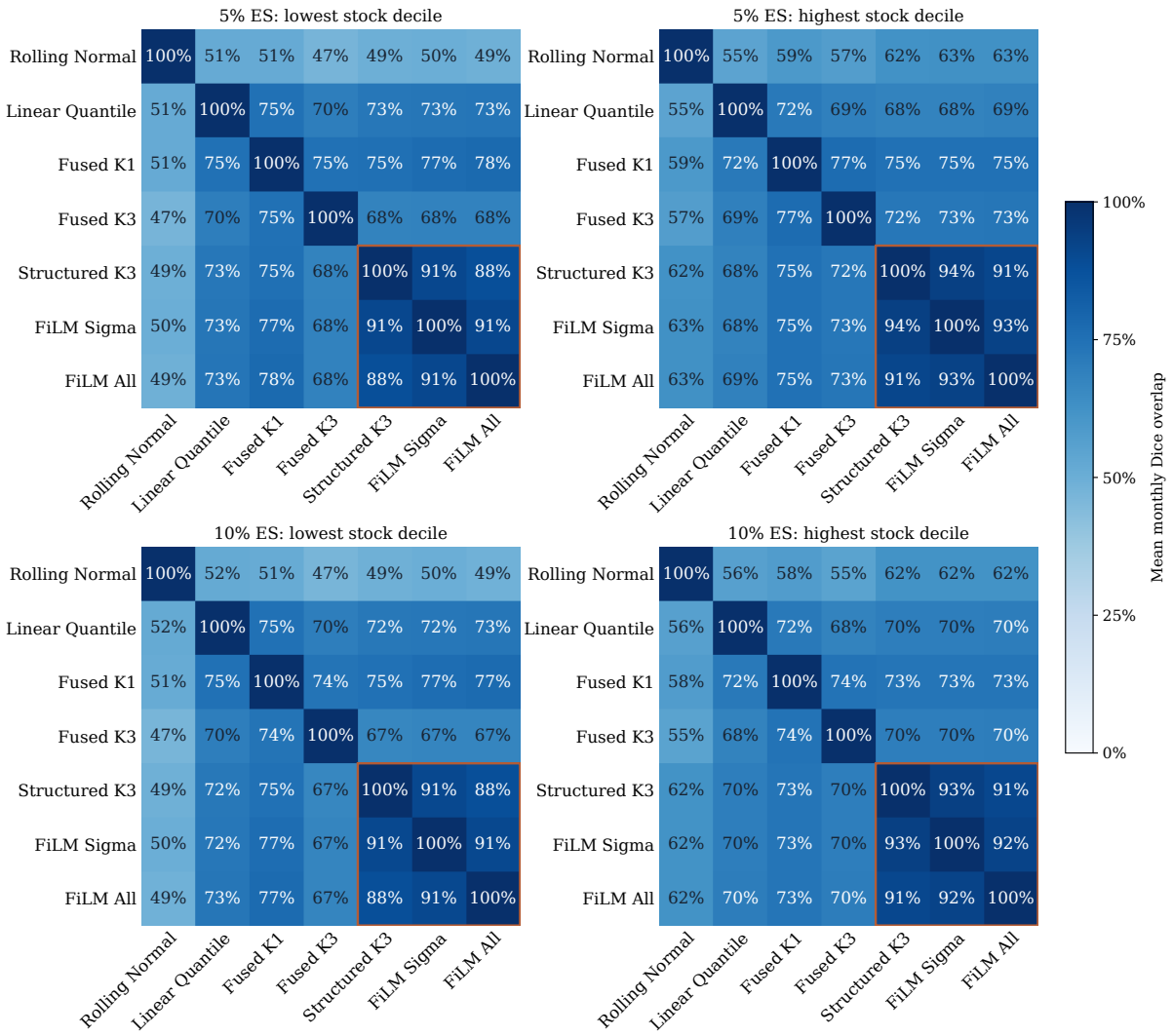


Figure 7: Overlap in extreme-ES stock selections

Notes: The seven-model matrices show average monthly Dice overlap for the lowest and highest stock deciles. The top row uses 5% ES and the bottom row uses 10% ES. All models use the same seven-model forecast universe of 480,446 stock–months over 216 months. This sample precedes the additional portfolio eligibility restrictions. The outlined block contains the three structured specifications. The diagonal is 100% by construction. Overlap describes selected stock membership, not equality of predicted ES values.

at both ES levels in the two additional validation seeds and the fixed-40-epoch run. Table E.6 reports those comparisons. Together with the forecast scores, these results show a consistent pattern across the three structured variants.

3.5 Distribution Shape Beyond the First Two Moments

A mixture can improve forecasts in two ways. It can predict the mean and volatility better, or it can describe features of the distribution that a Normal model misses, such as asymmetry. This exercise asks whether the second source of improvement matters. For each saved forecast from Structured K3, FiLM Sigma, and FiLM All, I construct a Normal forecast with exactly the same mean and variance:

$$\widetilde{F}^{N}_{i,t}(r) = \Phi\left(\frac{r - \widehat{m}_{i,t}}{\widehat{v}_{i,t}}\right), \tag{26}$$

Here $\widetilde{F}^{N}_{i,t}$ is the constructed Normal cumulative distribution function; the tilde marks the substitute and N stands for Normal. Its mean $\widehat{m}_{i,t}$ and variance $\widehat{v}^2_{i,t}$ come from the fitted mixture, using equation (3). No new model is trained. The mixture and its Normal substitute therefore predict exactly the same mean and volatility. Any difference in their scores comes from the rest of the distribution shape. Fused K1 remains a separate benchmark: it estimates its own mean and volatility, which can differ from those of the mixture.

I compare the two forecasts over all evaluation months and separately in bad markets. In addition to the distribution scores, I examine two outcomes that describe large stock losses. The first is whether a stock loses more than 10% next month:

$$D_{i,t+1} = \mathbf{1}\{R_{i,t+1} < -0.10\}. \tag{27}$$

The crash indicator $D_{i,t+1}$ equals one when the raw stock return is below -10% and zero otherwise. The second outcome measures how much the loss exceeds 10%:

$$L_{i,t+1} = \max\{-R_{i,t+1} - 0.10, 0\}. \tag{28}$$

For example, a return of -15% gives $L_{i,t+1} = 0.05$: the stock loses an additional five percentage points beyond the 10% threshold. A return of -8% gives zero. This realized excess loss uses a fixed threshold. ES instead averages predicted returns below the stock-specific VaR cutoff.

Bad markets are defined using the next-month market excess return. Let M_{t+1} be the

raw market return and $R_{f,t+1}$ the risk-free return. Their difference is $M^e_{t+1} = M_{t+1} - R_{f,t+1}$, where e means "excess." Let $c_{.20}$ be the 20th percentile of market excess returns across the 216 evaluation months. A month is classified as bad when its market excess return is at or below this cutoff:

$$B_{t+1} = \mathbf{1}\{M^e_{t+1} \le c_{.20}\}, \qquad c_{.20} = -0.0255. \tag{29}$$

The indicator B_{t+1} equals one in bad months and zero in other months. The cutoff is -2.55%, selecting 44 months with an average market excess return of -6.11%. The remaining 172 months are the other-market group. Forecasts, stock rankings, and portfolio positions are formed at month t. After returns are observed, this market classification identifies the months in which to evaluate their performance during market declines. The comparison asks whether information available before a decline distinguishes the stocks that subsequently suffer larger losses.

Each forecast distribution provides a crash probability, $\widehat{p}_{i,t}$, and an expected excess loss, $\widehat{\ell}_{i,t}$. Let $c = -0.10$ be the individual-stock crash threshold, separate from the market cutoff $c_{.20}$. For a mixture, the two predictions are:

$$\widehat{p}_{i,t} = \sum_k \pi_{k,t} \Phi\left(\frac{c - \mu_{i,k,t}}{\sigma_{i,k,t}}\right), \tag{30}$$

$$\widehat{\ell}_{i,t} = \sum_k \pi_{k,t} \left[(c - \mu_{i,k,t})\Phi\left(\frac{c - \mu_{i,k,t}}{\sigma_{i,k,t}}\right) + \sigma_{i,k,t}\phi\left(\frac{c - \mu_{i,k,t}}{\sigma_{i,k,t}}\right)\right]. \tag{31}$$

The Normal substitute uses the same formulas with one component. I evaluate crash probabilities using the Brier score, $(\widehat{p}_{i,t} - D_{i,t+1})^2$, and expected excess losses using squared error, $(\widehat{\ell}_{i,t} - L_{i,t+1})^2$. The average squared error is called mean squared error (MSE). I also compare NLL, exact CRPS, and the joint VaR–ES score. All comparisons use the same observations for the two forecasts. The reported difference is always the mixture loss minus the Normal substitute loss, so a negative number favors the mixture.

Table 8 shows two clear gains over the full sample. First, all three mixtures have lower NLL than their Normal substitutes: the differences are -0.01107 for Structured K3, -0.00790 for FiLM Sigma, and -0.01026 for FiLM All. All three differences are significant at 1%. The mixtures therefore assign more density near realized returns on average. Second, all three have lower MSE for expected losses beyond the -10% threshold, with p-values below 0.05. Because the competing forecasts have identical means and variances, these gains show that distribution shape adds useful information.

The clearest shape gains are in full-sample NLL and expected excess-loss forecasts. Full-sample CRPS and FZ0 differences at both ES levels are statistically insignificant. In

Table 8: Mixtures versus their own moment-matched Normal distributions

Model	Score	All: difference	p	Bad: difference	p
Structured K3	NLL	-0.011075	< 0.001	-0.002675	0.581
Structured K3	CRPS	0.000008	0.699	-0.000033	0.385
Structured K3	Excess-loss MSE	-0.000003	0.035	0.000010	0.010
Structured K3	FZ0 5%	-0.000747	0.645	-0.000402	0.957
Structured K3	FZ0 10%	-0.000320	0.688	0.004539	0.153
FiLM Sigma	NLL	-0.007899	0.002	-0.001990	0.547
FiLM Sigma	CRPS	-0.000010	0.602	-0.000068	0.310
FiLM Sigma	Excess-loss MSE	-0.000004	0.023	0.000014	0.017
FiLM Sigma	FZ0 5%	-0.000741	0.585	-0.004018	0.571
FiLM Sigma	FZ0 10%	-0.000493	0.503	0.001888	0.575
FiLM All	NLL	-0.010257	0.009	-0.004077	0.141
FiLM All	CRPS	0.000005	0.847	0.000016	0.813
FiLM All	Excess-loss MSE	-0.000004	0.033	0.000012	0.023
FiLM All	FZ0 5%	-0.002832	0.314	-0.018730	0.056
FiLM All	FZ0 10%	-0.001546	0.343	-0.005685	0.284

Notes: Differences are mixture loss minus the loss of its own moment-matched Normal substitute. Negative values favor the mixture. NLL, exact CRPS, and squared errors for losses beyond a raw return of -10% are independent of the ES probability level and appear once. FZ0 appears at both 5% and 10%, using the pairwise valid sample at each level. The history-complete sample has 410,049 observations; all and bad groups contain 216 and 44 months. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

bad months, FZ0 differences are also insignificant and excess-loss MSE is higher for the mixtures; crash-probability score differences vary across models. Thus, the full-sample log-score and excess-loss results identify the outcomes for which distribution shape adds predictive value.

3.6 Training Diagnostics

The forecast comparisons show what the fitted models predict. I next examine their training records to understand how these forecasts are obtained. I compare validation marginal NLL, a common measure of individual-stock density accuracy across all models. For the structured models, this is the marginal term recorded separately from the combined objective. Appendix E.1 provides the figures and cross-seed summaries.

The structured procedures show more sustained validation improvement and smaller typical epoch-to-epoch changes. In the main run, median validation marginal NLL improves relative to epoch 1 over the first 20 epochs for all three structured specifications, while it deteriorates for both fused benchmarks. Fused K1 and Fused K3 select epoch 1 in 10 and 11 of the 18 windows, respectively; the structured specifications never do so. Across all three validation seeds, each structured specification has smaller median absolute adjacent-epoch changes in validation marginal NLL over epochs 6–20 than either fused benchmark, and no epoch-1 selections. Checkpoints follow each model's validation objective, as defined in Section 2.2.

These patterns help interpret the forecast gains. The structured design restricts the probabilities to a common monthly function of macroeconomic information, while allowing firms to have different component means and scales. Its joint objective evaluates all firms under the same state, and training uses complete monthly cross-sections. This design links the treatment of common information to the way the model is fitted. The validation paths show more sustained improvement and smaller typical changes under the structured training procedures.

3.7 Main Forecast Results

The main forecast result is shared across Structured K3, FiLM Sigma, and FiLM All: each improves on the neural Normal and unrestricted mixture benchmarks. This finding remains when macroeconomic conditions affect only component probabilities or also adjust the firm features. Their extreme-ES stock selections are also much more similar to one another than to those of the fused benchmarks. The training diagnostics complement these forecast results with more sustained validation improvement under the structured

procedures. Section 4 first evaluates returns from total-ES sorts. Section 5 then asks whether component contributions provide additional information about stock losses, holding total ES and other risk measures fixed.

4 ES-Sorted Returns and Downside Protection

The forecast comparisons establish differences in predictive accuracy. This section asks whether sorting stocks by total ES also separates their realized returns, particularly in bad markets as defined in equation (29). Section 5 then studies the additional information in the component contributions.

4.1 Portfolio Formation and Factor Adjustment

Each month, I form two sets of portfolios by sorting eligible stocks separately on their original 5% and 10% ES. Each sort has ten groups. I suppress the tail-level index in the return equations below for readability. Breakpoints use the full eligible cross-section. Decile 1 has the most negative ES and the worst predicted tail loss. Decile 10 has the highest ES and the least severe predicted loss. The long–short return is

$$H_{t+1}^{ES} = R_{10,t+1} - R_{1,t+1}. \tag{32}$$

Here H_{t+1}^{ES} is the ES-sorted long–short portfolio return. The symbols $R_{10,t+1}$ and $R_{1,t+1}$ in this equation are returns on the tenth and first stock deciles, rather than returns on individual stocks. The superscript ES labels the sorting signal. The position buys the stocks with less severe predicted tail losses and sells those with more severe losses. I calculate both equal-weighted (EW) returns and value-weighted (VW) returns, with value weights given by formation-month market capitalization. Group membership, breakpoints, and weights are fixed before the next month's returns are observed. I hold the positions over the next month and use realized market returns to evaluate their performance in bad and other months. Returns and alphas are reported gross of trading and stock-borrowing costs, in monthly percentages. In tables, pp abbreviates percentage points, so a probability difference of 0.01 is one pp.

I evaluate full-period alphas using five factor models: CAPM, the Fama–French three-factor model (FF3), the Carhart four-factor model (Carhart4), FF5, and FF5 plus

momentum (FF5+MOM). The most complete specification is

$$H^{ES}_{t+1} = a + \beta_M M^e_{t+1} + \beta_S SMB_{t+1} + \beta_H HML_{t+1} + \beta_R RMW_{t+1} + \beta_C CMA_{t+1} + \beta_U UMD_{t+1} + u_{t+1}. \tag{33}$$

The factor returns SMB (small minus big), HML (high minus low book-to-market), RMW (robust minus weak profitability), and CMA (conservative minus aggressive investment) capture size, value, profitability, and investment. Together with M^e, they form the Fama–French five-factor model (FF5). The additional factor UMD (up minus down) captures momentum. The coefficients $\beta_M, \beta_S, \beta_H, \beta_R, \beta_C, \beta_U$ measure portfolio exposures to those six returns, and u_{t+1} is the regression residual. The intercept a is the monthly alpha in the displayed FF5+MOM regression (Fama and French, 2015; Carhart, 1997). The other specifications use subsets of these factor types: CAPM includes only the market; FF3 includes market, size, and value; Carhart4 adds momentum to FF3; and FF5 includes market, size, value, profitability, and investment. CAPM, FF3, and Carhart4 use the factor series from the FF3 data table; FF5 and FF5+MOM use those from the FF5 table. All five regressions use the same portfolio returns and months. Because the dependent variable is a long–short return, I do not subtract the risk-free rate again. Alpha estimates use all 216 months. Separately, I compare mean returns across bad and other markets. Mean-return and factor-regression tests use NW12 standard errors with calendar-month lags. Bad months retain their actual calendar dates instead of being treated as consecutive observations.

4.2 Returns Across Market States

Table 9 reports monthly mean returns for both ES levels and both weighting methods, separately for all, bad, and other months. At the 5% level, equal-weighted spreads earn 6.18%, 6.08%, and 5.96% per bad market month for Structured K3, FiLM Sigma, and FiLM All. At the 10% level, the corresponding returns are 6.23%, 6.10%, and 5.91%. All six means are significantly positive. Value-weighted bad-market spreads range from 5.21% to 5.89% at 5% ES and from 5.11% to 6.11% at 10% ES. In other months, equal-weighted spreads are slightly negative at both levels, ranging from −0.48% to −0.41%. Full-period equal-weighted means range from 0.84% to 0.91% and are not significant at 5%. The similar state pattern at both tail levels supports downside differentiation. Figure 8 shows both weighting schemes.

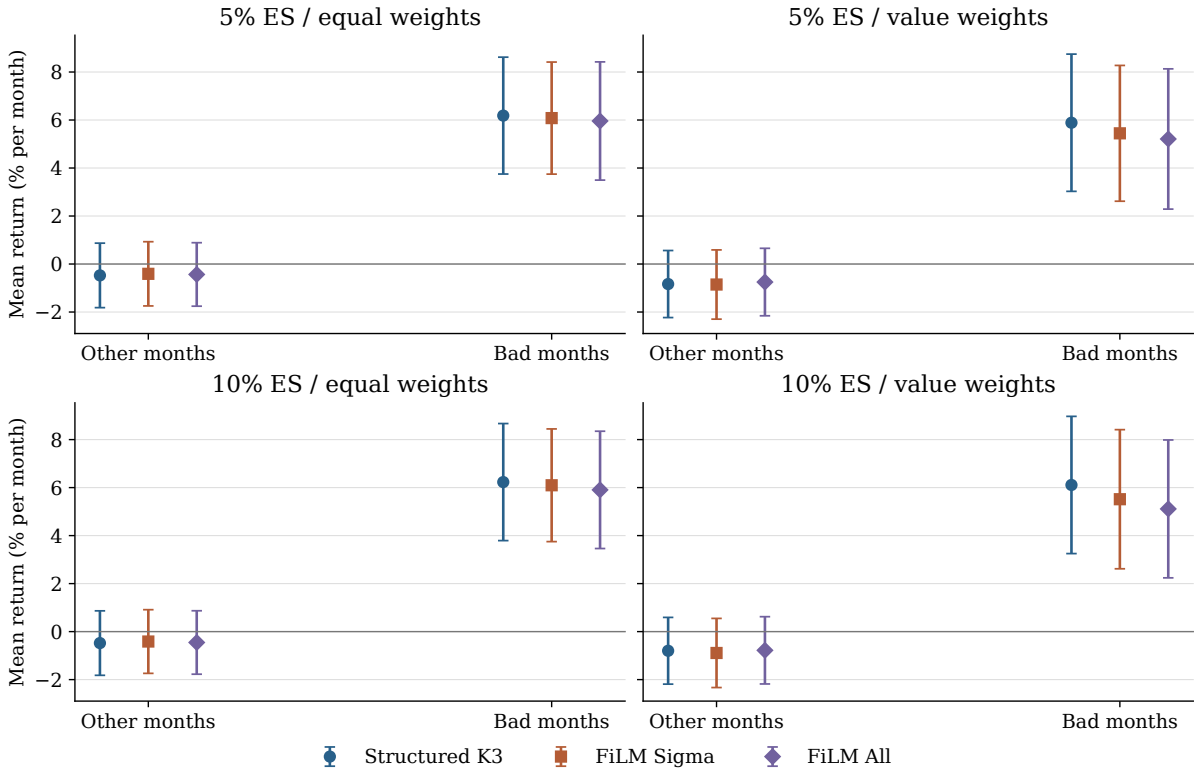


Figure 8: ES-sorted returns across market states and tail levels

Notes: Points are mean highest-minus-lowest ES stock-decile returns in monthly percentages. Rows use 5% and 10% ES; columns use equal and value weights. The same eligible stock sample and realized bad-market classification apply to both levels. Lines show pointwise 95% intervals based on calendar-time NW12 standard errors. Entries are gross mean return spreads.

Table 9: ES portfolio returns across market states

Model	Weight	All months	p	Bad months	p	Other months	p
Panel A: 5% ES							
Structured K3	EW	0.883	0.113	6.184	< 0.001	-0.474	0.490
Structured K3	VW	0.535	0.344	5.887	< 0.001	-0.834	0.242
FiLM Sigma	EW	0.914	0.104	6.080	< 0.001	-0.408	0.550
FiLM Sigma	VW	0.429	0.469	5.446	< 0.001	-0.854	0.246
FiLM All	EW	0.868	0.116	5.959	< 0.001	-0.434	0.520
FiLM All	VW	0.463	0.430	5.210	< 0.001	-0.751	0.295
Panel B: 10% ES							
Structured K3	EW	0.889	0.111	6.230	< 0.001	-0.478	0.485
Structured K3	VW	0.609	0.284	6.109	< 0.001	-0.799	0.260
FiLM Sigma	EW	0.913	0.102	6.097	< 0.001	-0.413	0.541
FiLM Sigma	VW	0.414	0.481	5.516	< 0.001	-0.891	0.226
FiLM All	EW	0.842	0.128	5.905	< 0.001	-0.453	0.502
FiLM All	VW	0.420	0.470	5.112	< 0.001	-0.780	0.275

Notes: Portfolios buy the highest-ES stock decile and sell the lowest-ES decile. Both ES levels use the same eligible stocks and separate ES rankings. EW gives stocks equal weights; VW uses formation-month market capitalization. Returns and alphas are monthly percentages. Tests use Newey–West standard errors with 12 calendar-month lags. All, bad, and other groups contain 216, 44, and 172 months. Bad markets are the bottom quintile of next-month market excess returns. The p-values test zero mean returns and are two-sided and unadjusted. Entries are gross mean return spreads.

4.3 Alphas Across Factor Models

Table 10 shows that the equal-weighted portfolio alpha is positive and significant at 5% under every factor model, for all three structured specifications and both ES levels. Across these six portfolios, monthly alphas range from 1.51% to 1.61% under CAPM, 1.30% to 1.39% under FF3, and 1.20% to 1.30% under Carhart4. They remain positive and significant under the more complete factor models: 0.76%–0.84% under FF5 and 0.67%–0.76% under FF5+MOM. Thus the positive equal-weighted alpha holds across five common choices of factor adjustment.

Value-weighted portfolios also have positive, significant alphas under CAPM, FF3, and Carhart4 at both ES levels. Their monthly alphas range from 1.11% to 1.36%, 0.87% to 1.13%, and 0.77% to 1.04%, respectively. Under FF5 and FF5+MOM, they fall to 0.25%–0.49% and 0.16%–0.42% and are not significant at 5%. The value-weighted alpha therefore depends more on the factor specification. Figure 9 shows these estimates and their confidence intervals.

ES forecasts identify stocks that perform better when the market falls. Buying stocks with higher ES and selling stocks with lower ES earns positive returns in bad markets

Table 10: ES portfolio alphas across factor models

Model	Weight	CAPM	FF3	Carhart4	FF5	FF5+MOM
Panel A: 5% ES						
		1.600***	1.384***	1.290***	0.835***	0.750***
Structured K3	EW	(3.45)	(4.27)	(4.22)	(2.82)	(2.77)
		1.274***	1.039**	0.953**	0.394	0.320
Structured K3	VW	(2.72)	(2.42)	(2.24)	(1.06)	(0.88)
		1.593***	1.373***	1.275***	0.815***	0.725***
FiLM Sigma	EW	(3.34)	(4.08)	(4.03)	(2.64)	(2.59)
		1.150**	0.923**	0.820**	0.265	0.172
FiLM Sigma	VW	(2.39)	(2.20)	(2.02)	(0.75)	(0.50)
		1.541***	1.324***	1.227***	0.781**	0.695**
FiLM All	EW	(3.26)	(3.95)	(3.88)	(2.52)	(2.45)
		1.157**	0.915**	0.816**	0.288	0.203
FiLM All	VW	(2.38)	(2.33)	(2.15)	(0.86)	(0.63)
Panel B: 10% ES						
		1.610***	1.393***	1.300***	0.842***	0.758***
Structured K3	EW	(3.49)	(4.36)	(4.32)	(2.89)	(2.87)
		1.363***	1.130***	1.045**	0.495	0.422
Structured K3	VW	(2.97)	(2.72)	(2.54)	(1.41)	(1.22)
		1.595***	1.374***	1.276***	0.815***	0.726***
FiLM Sigma	EW	(3.40)	(4.16)	(4.11)	(2.72)	(2.66)
		1.143**	0.916**	0.816**	0.265	0.175
FiLM Sigma	VW	(2.43)	(2.20)	(2.02)	(0.75)	(0.51)
		1.514***	1.297***	1.200***	0.759**	0.673**
FiLM All	EW	(3.19)	(3.87)	(3.80)	(2.47)	(2.39)
		1.111**	0.867**	0.770**	0.245	0.161
FiLM All	VW	(2.30)	(2.25)	(2.07)	(0.74)	(0.51)

Notes: Portfolios buy the highest-ES stock decile and sell the lowest-ES decile. Both ES levels use the same eligible stocks and separate ES rankings. EW gives stocks equal weights; VW uses formation-month market capitalization. Returns and alphas are monthly percentages. Tests use Newey–West standard errors with 12 calendar-month lags. Each regression uses all 216 months. Entries are intercepts with NW t-statistics in parentheses. CAPM uses the market factor; FF3 adds size and value; Carhart4 adds momentum to FF3; FF5 uses market, size, value, profitability, and investment; FF5+MOM adds momentum to FF5. CAPM, FF3, and Carhart4 use the FF3 factor series; FF5 and FF5+MOM use the FF5 series. *, **, and *** denote two-sided unadjusted $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

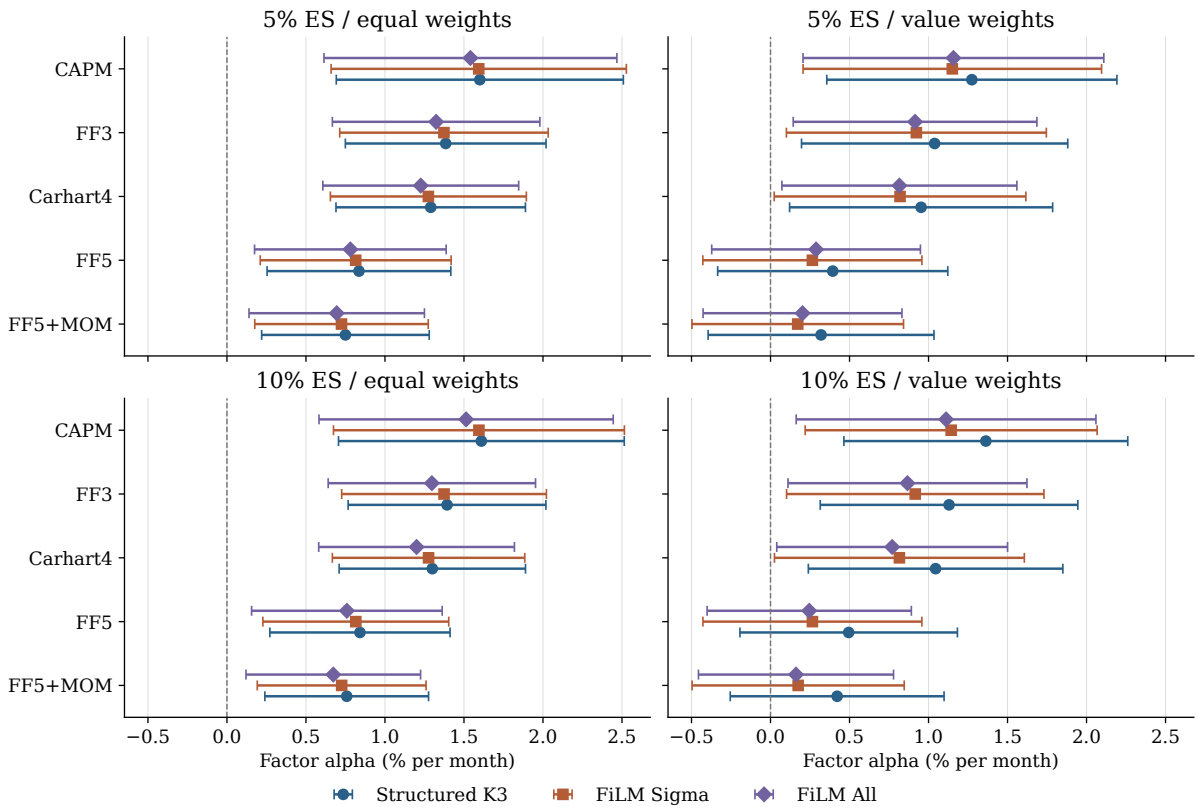


Figure 9: ES portfolio alphas across five factor models

Notes: Points are full-period monthly factor alphas for highest-minus-lowest ES stock-decile portfolios. Rows use 5% and 10% ES; columns use equal and value weights. Lines show pointwise 95% confidence intervals using NW12 standard errors. All specifications use the same 216 months and portfolio returns. The dashed line marks zero. These alphas are separate from the bad-market mean return spreads in Figure 8.

under both equal and value weighting. This result holds for all three structured models at both ES levels. The equal-weighted portfolios also earn positive and significant alphas under all five factor models. Total ES therefore provides useful information about downside performance and portfolio returns.

5 Tail-Risk Composition and Losses in Bad Markets

The preceding section shows that total ES distinguishes stocks with different downside performance. I now ask what the composition of that ES adds. Two stocks can have the same total ES while the low-mean component accounts for different amounts of their predicted losses. If these differences are related to subsequent crashes in bad markets, the decomposition provides useful information alongside the total risk measure. I test this relation with monthly regressions and residual contribution groups, controlling for total ES, historical exposures, and predicted mean and volatility.

5.1 Component Labels and Exact ES Decomposition

Changing component labels does not change the mixture density, provided that probabilities, means, and standard deviations are relabeled together. The same output channel can therefore have different meanings across estimation windows. For each window w, I assign labels using the selected model's fitted component means in the training and validation sample, as shown in Panel B of Figure 3. The adverse component is the one with the lowest average mean:

$$A_w = \arg\min_k \frac{1}{|\mathcal{A}_w|} \sum_{s \in \mathcal{A}_w} \frac{1}{N_s} \sum_i \mu_{i,k,s}, \tag{34}$$

The month set $\mathcal{A}_w$ is the 84 pretest training and validation months defined above. The index s runs over those months, N_s counts their firms, and $A_w \in \{1, 2, 3\}$ is the component index selected by arg min. The averaging first gives each firm equal weight within a month and then each month equal weight within $\mathcal{A}_w$. The component with the highest average mean is labeled favorable, and the remaining component is labeled middle. These labels remain fixed during the next 12 test months. Adverse therefore has a precise meaning: it is the component with the lowest average predicted return in the pretest sample. The following tests establish the economic information in its contribution by relating it to subsequent stock losses in bad markets.

Using ES from equation (4), all contributions remain in signed return units. Write $z_{i,k,t,\alpha} = (q_{i,t,\alpha} - \mu_{i,k,t})/\sigma_{i,k,t}$ for the VaR cutoff measured in component k's standard-deviation units. For each component, I calculate its share of outcomes in the mixture's

lower tail and its average return within that tail:

$$\omega_{i,k,t,\alpha} = \frac{\pi_{k,t}\Phi(z_{i,k,t,\alpha})}{\alpha}, \tag{35}$$

$$S_{i,k,t,\alpha} = \mu_{i,k,t} - \sigma_{i,k,t}\frac{\phi(z_{i,k,t,\alpha})}{\Phi(z_{i,k,t,\alpha})}. \tag{36}$$

Here $\omega_{i,k,t,\alpha}$ is the probability that a lower-tail observation comes from component k, conditional on falling below the mixture VaR. The monthly probability $\pi_{k,t}$ is common to all firms. The tail weight $\omega_{i,k,t,\alpha}$ can differ across firms because their component means, standard deviations, and VaR cutoffs differ. It measures the component's share of a particular stock's predicted tail outcomes. The quantity $S_{i,k,t,\alpha}$ is the component's mean return below that same cutoff; this indexed S is a tail severity, whereas S_{t+1} denotes the latent state. Their product is the additive contribution

$$C_{i,k,t,\alpha} = \omega_{i,k,t,\alpha}S_{i,k,t,\alpha} \tag{37}$$

$$= \frac{\pi_{k,t}}{\alpha}\left[\mu_{i,k,t}\Phi(z_{i,k,t,\alpha}) - \sigma_{i,k,t}\phi(z_{i,k,t,\alpha})\right]. \tag{38}$$

The symbol $C_{i,k,t,\alpha}$ therefore measures component k's signed contribution to total ES. Consequently,

$$\sum_k \omega_{i,k,t,\alpha} = 1, \qquad \mathrm{ES}_{i,t,\alpha} = \sum_k C_{i,k,t,\alpha}. \tag{39}$$

The adverse contribution is $C_{i,A_w,t,\alpha}$, written as $C^A_{i,t,\alpha}$. It is measured in signed return units. It is different from the component's overall probability $\pi_{A_w,t}$, its share of tail outcomes $\omega_{i,A_w,t,\alpha}$, and its average tail return $S_{i,A_w,t,\alpha}$. For example, a tail weight of 0.8 and an average tail return of -0.25 give a contribution of -0.20. If total ES is held fixed, a change in C^A must be offset by a change in the other components' combined contributions.

For month t, define the effective component count as $K_{\mathrm{eff},t} = 1/\sum_k \pi^2_{k,t}$. It approaches one when one component dominates and equals three for equal probabilities. Table 11 reports its equal-month average, denoted $\overline{K}_{\mathrm{eff}}$.

The component contributions add up to total ES with an error below 10^{-12} for all three models at both tail levels. The equal-month average effective component count ranges from 1.87 to 2.31. A component probability exceeds 0.95 in five months (2.31% of the sample) for Structured K3 and in no months for the two FiLM variants. The low-mean component accounts for 58.4%–63.5% of average 5% ES and 55.2%–60.1% of average 10% ES. These percentages measure ES contributions, not state probabilities. The component labeled adverse using pretest data remains the lowest-mean component for more than 99.5% of firms in the average test month for each model.

Table 11: Use and numerical consistency of the aligned components

Model	$\overline{K}_{\text{eff}}$	Collapse (%)	Lowest mean (%)	ES share 5%	ES share 10%	Max ES error
Structured K3	1.870	2.31	99.92	58.70	56.30	4.4×10^{-16}
FiLM Sigma	2.159	0.00	99.57	58.42	55.19	3.3×10^{-16}
FiLM All	2.306	0.00	99.63	63.53	60.05	4.4×10^{-16}

Notes: The full-distribution diagnostics appear once. Effective components are $1/\sum_k \pi_k^2$; collapse means a probability above 0.95. Lowest mean is the equal-month average fraction of firms for which the pretest low-mean label stays lowest. Each ES share is 100 times the equal-month average low-mean contribution divided by average total ES at that level, not a state probability. Max ES error is the largest absolute reconstruction error across both levels.

5.2 Historical Exposures and Forecast Controls

The primary outcome is the individual crash indicator in equation (27); additional tests use raw returns and excess loss from equation (28). Bad markets follow equation (29). The controls below ask whether contribution adds information beyond total ES and measured historical exposures.

The history controls are log size, the trailing 12-month mean return and volatility, and momentum that skips the formation month. I then add market beta, downside beta, and market-model residual volatility. These exposures use the 60 calendar months ending at t. They require at least 36 valid stock–market excess-return pairs and at least 12 downside pairs. A downside month has a market excess return below the market's mean over that window. Missing betas remain missing, so stocks without enough history are excluded from the regression sample. The final specification also controls for the MDN's predicted mean and standard deviation from equation (3). Appendix A.3 gives the historical exposure formulas and observation requirements.

5.3 Monthly Cross-Sectional Regressions

Total ES measures the size of a stock's predicted tail loss. The adverse contribution measures how much of that loss comes from the component with the lowest pretest mean. This section asks whether knowing the source of the loss adds information: after controlling for total ES and other risk measures, does a lower adverse contribution identify stocks that are more likely to crash in bad markets?

I answer this question in two steps. First, I estimate the relation between contribution and next-month stock outcomes separately in each formation month. This produces one contribution coefficient per month. Second, I compare these coefficients between bad and other market months. This separates the comparison across stocks from the comparison

across market conditions.

5.3.1 Total ES and Stock Losses

I first check whether total ES alone is related to realized losses. Each monthly regression includes total ES, historical size and return controls, and industry effects, but omits the contribution. In bad markets, the ES coefficients for crash probability are -0.717, -0.685, and -0.722 at the 5% ES level for Structured K3, FiLM Sigma, and FiLM All. At the 10% level, they are -0.823, -0.792, and -0.831. All six have $p < 0.001$.

These numbers have a direct interpretation. An increase in ES from -0.30 to -0.29 means that the predicted tail loss becomes one percentage point smaller. Such an increase is associated with roughly 0.7–0.8 percentage points lower crash probability in bad months, holding the included controls fixed. Higher ES is also associated with smaller losses beyond the -10% crash threshold. Thus total ES already contains useful information about subsequent stock losses. Table C.1 in Appendix C reports these estimates for all, bad, and other months. The next question is whether the adverse contribution adds information beyond this relation with total ES.

5.3.2 Step 1: Compare Stocks Within Each Month

For each model and ES level, I add the adverse contribution to the monthly regression. Let $Y_{i,t+1}$ be stock i's outcome in the next month. The main outcome is the crash indicator $D_{i,t+1}$, equal to one when the raw return is below -10%. I also use raw return $R_{i,t+1}$ and excess loss $L_{i,t+1}$ as outcomes. The index s identifies one of the four control specifications, and $X_{i,t}^{(s)}$ is the corresponding vector of ordinary controls. I estimate

$$\begin{aligned} Y_{i,t+1} = a_t^{(s)} + b_{E,t}^{(s)} E_{i,t,\alpha}^{W} + b_{C,t}^{(s)} C_{i,t,\alpha}^{A,W} + \boldsymbol{\theta}_t^{(s)\prime} Z_t^{W}(X_{i,t}^{(s)}) \\ + \sum_{j \neq j_0} \eta_{j,t}^{(s)} \mathbf{1}\{\text{industry}_{i,t} = j\} + \varepsilon_{i,t+1}^{(s)}. \end{aligned} \tag{40}$$

Here $E_{i,t,\alpha} = \text{ES}_{i,t,\alpha}$ is total ES and $C_{i,t,\alpha}^{A}$ is the adverse component's contribution at the same tail level α. The superscript A identifies that component. Both variables retain their signed decimal return units. A lower contribution means a larger predicted loss from the adverse component. Holding total ES fixed, this larger adverse loss must be offset by a smaller combined loss, or a larger gain, from the other components.

The coefficient of interest is $b_{C,t}^{(s)}$. It measures how next-month outcomes differ with contribution after accounting for total ES and the controls. With crash probability as the outcome, a negative coefficient means that a lower contribution is associated with a higher

crash probability. The coefficient $b_{E,t}^{(s)}$ measures the relation with total ES while holding contribution and the controls fixed. These are different comparisons: the first concerns the source of the tail loss, while the second concerns its total size.

The units are also straightforward. A one-percentage-point increase in contribution, such as -0.20 to -0.19, corresponds to $b_{C,t}^{(s)}$ percentage points change in fitted crash probability. For example, if the coefficient is -2, that increase is associated with a two-percentage-point lower crash probability. This example explains the scale of the coefficient; it is not an estimated value from the table below.

The remaining terms account for other differences across stocks. The coefficient $a_t^{(s)}$ is the monthly intercept, $\boldsymbol{\theta}_t^{(s)}$ contains the ordinary-control coefficients, and $\varepsilon_{i,t+1}^{(s)}$ is the unexplained outcome. A prime denotes transpose, so the product with the control vector sums each control times its coefficient. The industry indicator equals one when stock i belongs to industry j. The reference industry is j_0, and $\eta_{j,t}^{(s)}$ is industry j's adjustment relative to that reference. Industry effects allow average outcomes to differ across industries. Here j indexes industries; it is separate from the predictor index used in the input-scaling formula. All coefficients are estimated anew each month.

The transformations in equation (40) apply only to these regressions. The superscript W means that observations below the monthly 1st percentile or above the 99th percentile are set to those cutoffs. ES and contribution are kept in their resulting decimal return units. For ordinary controls, Z_t^W applies the same winsorization and then subtracts the monthly cross-sectional mean and divides by the monthly cross-sectional standard deviation. The regression vector $X_{i,t}^{(s)}$ is therefore distinct from the neural input $x_{i,t}$, and this within-month transformation differs from the training-window scaling in equation (11). Realized outcomes are not winsorized.

5.3.3 Step 2: Compare Bad and Other Market Months

The first step gives 216 monthly contribution coefficients, denoted by $\widehat{b}_{C,t}^{(s)}$. The hat marks an estimated coefficient. I next ask whether these coefficients become more negative in bad markets. Let $w(t)$ identify the annual estimation window used for month t. I estimate

$$\widehat{b}_{C,t}^{(s)} = \kappa^{(s)} + \delta_C^{(s)} B_{t+1} + \sum_{w=1}^{17} \lambda_w^{(s)} \mathbf{1}\{w(t) = w\} + u_t^{(s)}. \tag{41}$$

The outcome in this regression is the monthly contribution coefficient from the first step. The indicator B_{t+1} equals one when next month's market excess return is in the bottom quintile, as defined in equation (29). It selects 44 bad months; the remaining 172 months

are the other-market group. The second-stage intercept is $\kappa^{(s)}$, and $u_t^{(s)}$ is the monthly regression residual.

The coefficient $\delta_C^{(s)}$ is the central result: it measures how much the contribution coefficient changes in bad months relative to other months. A negative value means that the relation between a lower contribution and higher crash probability is stronger in bad markets. It is a difference between slopes, not the average crash rate or the contribution coefficient in bad months alone.

The window indicators account for changes between the 18 annually fitted models. The coefficient $\lambda_w^{(s)}$ allows each window to have a different average contribution slope. Window $w = 0$ is the reference, so the regression includes indicators for $w = 1, \ldots, 17$. This adjustment prevents differences in average slopes across annual estimation windows from being treated as a bad-market effect. Each of the 216 monthly coefficients receives equal weight in this regression. NW12 standard errors allow heteroskedasticity and dependence across calendar months in its monthly regression errors. Appendix B.4 gives the calendar-time covariance calculation.

5.3.4 Results After Controlling for Other Risk Measures

Table 12 reports δ_C as the control set expands. The first specification controls for size, historical mean return, historical volatility, momentum, and industry. The second adds historical market beta. The third also adds downside beta and idiosyncratic volatility. The fourth adds the model's predicted mean and standard deviation. Every specification includes total ES and adverse contribution. All use the same 410,049 stock–months, so the changes between columns are not caused by changing the stock sample.

The result holds across all four control sets: all 24 estimates are negative and significant at 5%, covering three structured models, two ES levels, and four specifications. Adding historical market and downside exposures does not remove the relation. Including predicted mean and volatility does not remove it either. Thus the adverse contribution contains information about bad-market crash vulnerability beyond total ES and the included controls.

With all controls included, the bad-minus-other slope differences are -4.114, -2.386, and -2.343 at 5% ES for Structured K3, FiLM Sigma, and FiLM All. At 10% ES, they are -6.239, -3.767, and -3.495. All six p-values are below 0.001. The common result across the three specifications matters more here than their numerical ordering: under each structured design, a lower adverse contribution is more closely associated with crashes when the market performs poorly.

For a concrete interpretation, consider the Structured K3 estimate of -4.114 at 5%

Table 12: Tail contribution and bad-market crash sensitivity at both ES levels

Model	Controls	δ_C: 5%	SE	p	δ_C: 10%	SE	p
Structured K3	1 History	-4.240	1.212	< 0.001	-6.319	1.686	< 0.001
Structured K3	2 + market beta	-3.542	1.117	0.002	-5.290	1.563	< 0.001
Structured K3	3 + downside / residual volatility	-3.573	1.119	0.001	-5.342	1.563	< 0.001
Structured K3	4 + predicted moments	-4.114	1.227	< 0.001	-6.239	1.883	< 0.001
FiLM Sigma	1 History	-2.454	0.350	< 0.001	-3.644	0.531	< 0.001
FiLM Sigma	2 + market beta	-2.005	0.315	< 0.001	-3.002	0.482	< 0.001
FiLM Sigma	3 + downside / residual volatility	-2.018	0.299	< 0.001	-3.028	0.457	< 0.001
FiLM Sigma	4 + predicted moments	-2.386	0.345	< 0.001	-3.767	0.544	< 0.001
FiLM All	1 History	-2.314	0.381	< 0.001	-3.398	0.583	< 0.001
FiLM All	2 + market beta	-1.869	0.339	< 0.001	-2.744	0.513	< 0.001
FiLM All	3 + downside / residual volatility	-1.892	0.326	< 0.001	-2.785	0.496	< 0.001
FiLM All	4 + predicted moments	-2.343	0.363	< 0.001	-3.495	0.580	< 0.001

Notes: Both tail levels use the same 410,049 stock–months and 216 months. Each first stage includes total ES and its low-mean component contribution at the indicated tail level, plus industry effects and the listed controls. History includes log size, historical mean and volatility, and momentum. The second stage compares monthly contribution slopes between bad and other markets, controlling for estimation-window effects. Units are crash-probability percentage points per one-percentage-point increase in signed contribution. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

ES. In the regression, compare a contribution of -0.21 with -0.20, keeping total ES and all controls fixed. The stock with the lower contribution has a larger predicted loss from the adverse component. The crash-probability difference associated with this one-percentage-point lower contribution is 4.114 percentage points larger in bad months than in other months, after the window adjustment. The estimate measures how much the difference between the stocks increases in bad markets relative to other months.

Contribution signals have different cross-sectional scales across models and ES levels. The next subsection therefore compares actual stock groups, providing an additional measure of economic magnitude.

5.3.5 Returns and the Size of Losses

The finding also appears when the outcome changes. With raw return as the outcome and all controls included, the bad-minus-other contribution slopes are positive for all three models. They range from 0.455 to 0.853 at 5% ES and from 0.679 to 1.272 at 10% ES; all six are significant at 1%. Thus a lower contribution is associated with an additional decline in returns in bad markets, relative to its relation with returns in other months.

The excess-loss results point in the same direction. Recall that excess loss is the amount by which a realized stock loss exceeds 10%. The slope differences are negative for all three models, ranging from -0.317 to -0.165 at 5% ES and from -0.489 to -0.274 at 10%

ES. All six are significant at 5%. A lower contribution is therefore associated with larger additional losses beyond the crash threshold in bad months. In both outcome regressions, units are outcome percentage points per one-percentage-point increase in contribution. Table C.2 in Appendix C reports the model-specific estimates.

These regressions give a consistent interpretation of the decomposition. Total ES identifies stocks with different overall downside exposure. After total ES and the measured controls are included, the adverse contribution still distinguishes stocks whose crash probabilities rise more, whose returns fall more, and whose losses extend further below the crash threshold in bad markets. The result holds for all three structured models at both ES levels. The following residual sorts make these differences easier to see in groups of stocks.

5.4 Residual Contribution Groups and Crash Rates

The preceding regressions show how crash probability changes with contribution. I now compare two groups of stocks to show the size of the difference in realized outcomes. The question is whether stocks whose adverse contribution is lower than total ES and other characteristics would suggest have higher crash rates, especially in bad markets.

5.4.1 Forming the Two Stock Groups

Directly sorting on adverse contribution could also separate stocks by total ES, historical exposures, or predicted volatility. To focus on the information left after accounting for those variables, I first regress adverse contribution on total ES, the full control set, and industry indicators within each formation month:

$$C^{A,W}_{i,t,\alpha} = a_t + \rho_t E^W_{i,t,\alpha} + \boldsymbol{\psi}_t' Z^W_t(X^{\text{full}}_{i,t}) + \sum_{j \neq j_0} \xi_{j,t} \mathbf{1}\{\text{industry}_{i,t} = j\} + \widetilde{C}^A_{i,t,\alpha}. \tag{42}$$

The variable on the left is the model's adverse contribution for stock i at month t and ES level α. On the right, a_t is the intercept, ρ_t is the coefficient on total ES, $\boldsymbol{\psi}_t$ contains the ordinary-control coefficients, and $\xi_{j,t}$ is the effect of industry j relative to the reference industry j_0. The superscript full selects the final control set, including historical exposures and the predicted mean and volatility. As in the preceding regressions, W denotes monthly winsorization and Z^W_t additionally standardizes the ordinary controls within month.

The residual $\widetilde{C}^A_{i,t,\alpha}$ is the model's adverse contribution minus the fitted value from this monthly regression. It is the part left after removing the fitted linear relation with total ES and the listed controls. A negative residual means that the contribution is more negative

than the linear fit based on total ES and the controls. A positive residual means that it is less negative, or more positive, than that fit. For example, if the fitted contribution is -0.20 and the model's contribution is -0.24, the residual is -0.04. The model assigns four percentage points more loss to the adverse component than the fitted relation would suggest. Both contributions in this calculation are based on information at t; neither is a realized next-month loss. This residual is also different from a stock-return residual.

Each month, I rank stocks from the lowest to the highest contribution residual and divide them into ten groups. I compare the bottom 10% with the top 10%. The bottom group has the lowest residuals: its adverse contribution is relatively low after the linear adjustment. The top group has the highest residuals. This procedure is repeated separately for each of the three structured models and for 5% and 10% ES, giving six sets of monthly sorts. Group membership is determined before observing the next month's returns or market state. The 5% and 10% ES levels determine the contribution signal; the two stock groups each contain approximately 10% of stocks in either case.

5.4.2 Measuring the Difference in Crash Rates

After the next month's returns are observed, I calculate the fraction of stocks in each group that lose more than 10%. I then subtract the top group's crash rate from the bottom group's crash rate:

$$G_{t+1} = \frac{1}{N_{1,t}} \sum_{i \in \mathcal{D}_{1,t}} D_{i,t+1} - \frac{1}{N_{10,t}} \sum_{i \in \mathcal{D}_{10,t}} D_{i,t+1}. \tag{43}$$

Here $\mathcal{D}_{1,t}$ and $\mathcal{D}_{10,t}$ are the bottom and top residual stock groups, and $N_{1,t}$ and $N_{10,t}$ are their numbers of stocks. The indicator $D_{i,t+1}$ is one when the stock return is below -10% and zero otherwise. Thus each sum divided by its group size is a crash fraction. The model and ES-level indices are omitted from G_{t+1} for readability, but each comparison uses its own groups.

A positive G_{t+1} means that more stocks crash in the bottom residual group. Multiplying G_{t+1} by 100 expresses the difference in percentage points. For illustration, crash rates of 30% and 23% would give $G_{t+1} = 0.07$, or a seven-percentage-point gap. This is a difference between two crash rates, not a portfolio return or the crash rate of either group alone.

I calculate one gap for each month and average these gaps separately over the 44 bad months and the 172 other months. Each month receives equal weight. Bad months are identified afterward using the bottom quintile of next-month market excess returns; this label does not enter the stock ranking. In Table 13, "Bad gap" and "Other gap" report these two averages.

The table also reports "Adjusted Δ." To calculate it, I regress the monthly crash-rate gap G_{t+1} on the bad-market indicator and annual estimation-window indicators. The coefficient on the bad-market indicator measures how much larger the gap is in bad months than in other months, after allowing each training window to have its own average gap. It need not equal the simple difference between the Bad gap and Other gap columns, because it also controls for these window differences. All tests and confidence intervals use calendar-time NW12 standard errors.

5.4.3 Results and Their Graphical Interpretation

Table 13: Residual contribution sorts at the 5% and 10% ES levels

Model	ES (%)	Bad gap	p	Other gap	p	Adjusted Δ	p
Structured K3	5	6.449	< 0.001	0.091	0.836	6.710	< 0.001
Structured K3	10	6.463	< 0.001	-0.003	0.994	6.757	< 0.001
FiLM Sigma	5	7.141	< 0.001	-0.406	0.294	7.944	< 0.001
FiLM Sigma	10	7.596	< 0.001	-0.205	0.624	8.202	< 0.001
FiLM All	5	6.588	< 0.001	0.093	0.806	6.817	< 0.001
FiLM All	10	6.207	< 0.001	0.162	0.656	6.343	< 0.001

Notes: At each tail level, contribution is residualized monthly on total ES at the same level, all controls, and industry effects. The gap is the lowest-minus-highest residual decile difference in next-month crash rates, in percentage points. The stock crash cutoff remains a raw return below −10%; bad markets remain the same 44 months. Adjusted Δ controls for estimation-window effects. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

The bottom residual group has a substantially higher crash rate in bad markets. At 5% ES, the bad-month gaps are 6.45 percentage points for Structured K3, 7.14 for FiLM Sigma, and 6.59 for FiLM All. At 10% ES, they are 6.46, 7.60, and 6.21 points. All six gaps have $p < 0.001$. These are sizeable differences in the fraction of stocks that actually lose more than 10% in the following month.

In other months, the gaps are between −0.41 and 0.16 percentage points, and none is statistically different from zero. The clear separation between the two stock groups therefore appears in bad markets. The window-adjusted comparisons confirm this pattern: Adjusted Δ ranges from 6.34 to 8.20 percentage points across the six model–ES combinations, with $p < 0.001$ in every case.

Figure 10 plots the Bad gap and Other gap columns of Table 13. The left panel uses 5% ES contributions and the right panel uses 10% ES contributions. In each panel, the three markers identify Structured K3, FiLM Sigma, and FiLM All. Their heights show the average bottom-minus-top crash-rate gap in percentage points, and the vertical bars

show pointwise 95% confidence intervals. For example, the blue bad-month point in the left panel is 6.45: the bottom residual group's crash rate exceeds the top group's by 6.45 percentage points on average in bad months.

The figure makes the main pattern visible. In other months, all three estimates are close to zero and their intervals cross zero. In bad months, all three estimates are well above zero and their intervals remain above zero. Both ES levels show the same pattern. These plotted points are the average group gaps before the annual-window adjustment; the adjusted bad-minus-other differences appear only in the table.

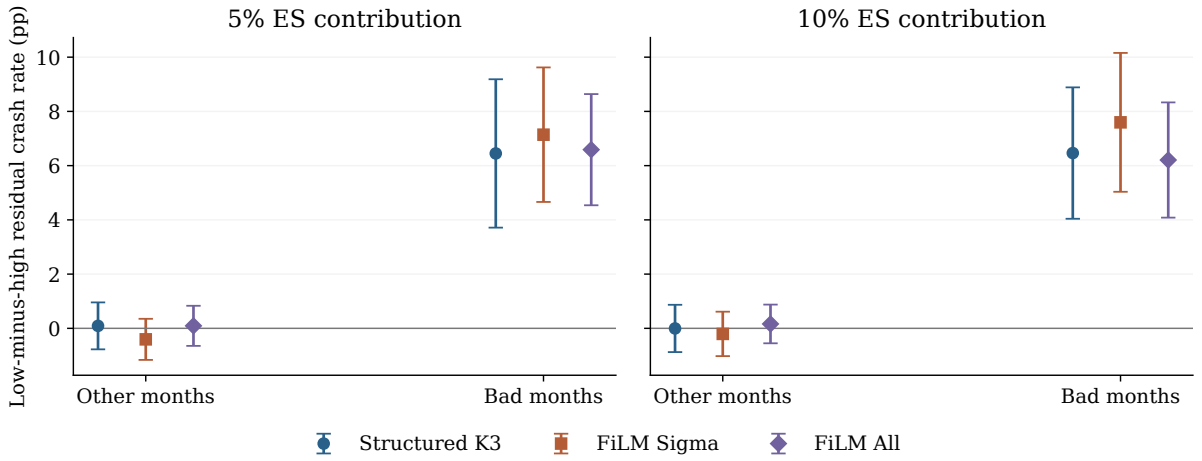


Figure 10: Crash-rate differences between low and high contribution-residual groups
Notes: Each month, I regress adverse contribution on total ES, all controls, and industry indicators, and calculate each stock's residual as contribution minus its fitted value. Before observing next month's returns, I rank these residuals and select the bottom 10% and top 10% of stocks. Each point shows the bottom group's next-month crash rate minus the top group's rate, averaged over bad or other months with equal monthly weights. A crash is a raw stock return below −10%. The left panel uses 5% ES contributions and the right panel uses 10% ES contributions. Bad markets follow equation (29). Vertical bars are pointwise 95% confidence intervals based on calendar-time NW12 standard errors. Units are percentage points. Points correspond to the Bad gap and Other gap table entries, not Adjusted Δ.

The residual ranking uses the part of contribution left after its monthly linear adjustment for total ES and the controls. Table C.4 in Appendix C reports the ES, beta, and predicted-volatility differences between the selected groups, while Table C.3 reports the variation retained after adjustment. The regressions and group comparisons provide two ways to describe the additional information in contribution.

The economic finding is that the source of predicted tail losses matters alongside their total size. The part of adverse contribution left after the linear adjustment identifies groups with roughly six to eight percentage points different crash rates in bad markets, while their other-month gaps are small. This result holds for all three structured models at both ES levels and makes the conditional relation in the preceding regressions visible in

actual stock groups.

6 Robustness and Training Sensitivity

The baseline uses seed 2027 and validation-based checkpoint selection.[1] The two additional validation runs use seeds 2026 and 42. The fixed-epoch check also uses seed 2027. All four runs include the formation-month return among the 21 firm inputs. They use the same forecast period, market-state definition, and empirical testing procedures. Appendix E reports each run separately, including its fused benchmarks.

A fixed market excess-return cutoff of -5% selects 25 months. With all controls included, contribution differences remain negative and significant at 5% for all three models. A -3% cutoff selects 40 months and gives the same conclusion. The side-by-side results above establish the contribution finding at both ES levels. Results at both ES levels are significant with 3, 6, or 12 HAC lags. Appendix D reports these checks; the main market rule remains the bottom quintile and the main lag length remains 12 months.

Across all nine model–seed combinations under validation selection, the structured models have lower reported density, distribution, and tail losses than both fused benchmarks. The fully controlled contribution difference is always negative and significant. Bad-market residual crash gaps range from 3.58 to 7.14 percentage points at 5% ES and from 3.73 to 7.60 points at 10% ES; all are significant at 1%. At 5% ES, none of the other-month gaps is significant at 5%. At 10%, one of nine other-month gaps is significant and negative, while the other eight are insignificant. Across fits, the contribution groups consistently differ more in bad markets. This pattern holds for all three structured variants. Their extreme-ES selections also remain more similar within the structured family than across structured–fused pairs in every run (Table E.6).

The fixed-40-epoch run combines training and validation into 84 training months, uses a constant learning rate, and keeps the final checkpoint. All three models continue to improve on the fused benchmarks and have negative, significant contribution differences. Their bad-market residual crash gaps range from 5.68 to 7.03 percentage points. Window-adjusted bad-minus-other gaps are also positive and significant for all three models. FiLM All also has a smaller significant gap in other months. Across the three models, the larger separation occurs in bad markets.

These checks evaluate sensitivity to initialization and training protocol using the same stock returns. Each run has its own reported results and statistical tests. Appendix E

[1] Seed 2027 was selected after comparing the available runs during analysis. The additional seeds and training protocol are reported separately in Appendix E.

also reports their distribution-shape comparisons and portfolio alphas.

7 Conclusion

This paper shows the value of giving macroeconomic and firm information distinct roles in return-distribution forecasting. Macroeconomic inputs determine probabilities shared by all firms in a month, while firm characteristics determine the return distribution within each component. The baseline and both FiLM extensions improve on the neural Normal and unrestricted mixture benchmarks. Their forecast gains persist across the additional seeds and training protocol, and their extreme-ES selections are closely aligned. The recurring finding is the value of the common structured design across its three implementations.

The forecasts also distinguish economically important differences across stocks. Portfolios formed before returns are observed earn positive gross spreads during market declines when they buy stocks with less severe predicted tail losses and sell stocks with more severe losses. This pattern holds under both equal and value weighting and at both ES levels. Equal-weighted portfolios also have positive, significant gross alphas under all five factor models.

The ES decomposition adds a second source of information. After accounting for total ES, historical exposures, and predicted mean and volatility, stocks with lower residual adverse contributions have substantially higher crash rates in bad markets. The corresponding differences are small in other months in the main specification, and the concentration of larger gaps in bad markets persists across the alternative fits. The component contribution therefore helps distinguish vulnerability to market declines after the monthly linear adjustment for total ES and the controls.

For risk measurement, the implication is direct: both the size and the composition of a predicted tail loss are informative. Sharing monthly probabilities provides a coherent way to combine a common market environment with firm-level differences. The resulting forecasts improve statistical accuracy and reveal differences in the stock losses that occur when markets fall.

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A Data, Predictors, and Regression Controls

A.1 Firm Data

The models use 21 continuous firm inputs: 15 accounting ratios, five derived market variables, and the raw formation-month return, `mthret`. The last variable is observed in month t and predicts the return in $t + 1$. Price is used in sample eligibility and variable construction but is not a separate continuous neural input. I first build the calendar grid and next-month target, then restrict formation months to 2000–2024. Only formation months in 2007–2024 are used for out-of-sample evaluation.

Table A.1: Continuous firm predictors

Field	Description
`bm`	Book-to-market
`evm`	Enterprise-value multiple
`ps`	Price-to-sales
`pcf`	Price-to-cash-flow
`opmad`	Operating profit margin after depreciation
`cfm`	Cash-flow margin
`roe`	Return on equity
`gprof`	Gross profitability
`debt_at`	Debt-to-assets
`short_debt`	Short-term debt share
`cash_ratio`	Cash ratio
`curr_ratio`	Current ratio
`at_turn`	Asset turnover
`rd_sale`	Research and development to sales
`accrual`	Accruals
`MOM12m`	Compounded returns over months $t-11$ through $t-1$; at least six valid months
`mu_bench`	Mean return over months $t-11$ through t; at least six valid months
`sigma_bench`	Sample return standard deviation over months $t-11$ through t; at least six valid months
`turnover`	Monthly volume divided by shares outstanding (in shares)
`mthcap_log`	Log formation-month market capitalization
`mthret`	Raw return in formation month t; predicts the return in month $t+1$

Notes: Accounting ratios follow the WRDS definitions. Market variables use the complete CRSP calendar panel. Turnover divides monthly volume by stored shares outstanding multiplied by 1,000 to obtain shares. Industry enters separately as a learned embedding, with a category for missing classifications.

Accounting ratios are linked to CRSP firms through the Compustat link and recorded public date, then carried forward to the formation month. The date check confirms that the recorded public date is no later than formation. It does not reconstruct revisions that are absent from the stored data. The five derived predictors are calculated before price and industry filters, so removing a stock from the sample does not compress its calendar history.

A.2 Macro Data

Table A.2: Macroeconomic and financial information

Series	Source	Description
PAYEMS	FRED-MD	Nonfarm employment; vintage transformation
UNRATE	FRED-MD	Unemployment rate
INDPRO	FRED-MD	Industrial production; vintage transformation
RPI	FRED-MD	Real personal income; vintage transformation
ISRATIOx	FRED-MD	Inventories-to-sales ratio; vintage transformation
HOUST	FRED-MD	Housing starts; vintage transformation
BUSLOANS	FRED-MD	Business loans; vintage transformation
CPIAUCSL	FRED-MD	Consumer price index; vintage transformation
VIXCLS	FRED	Monthly average VIX
TERM_SPREAD	FRED	10-year Treasury yield minus 3-month bill yield
CREDIT_SPREAD	FRED	Baa corporate yield minus 10-year Treasury yield
FEDFUNDS	FRED	Monthly change in averaged effective policy rate
sprtrn	WRDS	S&P 500 monthly return

Notes: FRED-MD transformations are applied within each forecast-month vintage. The eight FRED-MD series use their specified transformations where required. Other inputs are measured monthly. All 13 variables are scaled using one observation per training month.

Each forecast month has a corresponding FRED-MD vintage file. I apply the supplied transformation code within that file and select the latest nonmissing transformed value for each series. The observation month can therefore differ across series in the same vintage. Additional financial series use monthly averages of daily FRED values. The policy-rate variable is the monthly change in the averaged effective federal funds rate, and the equity-market variable is the S&P 500 monthly return. Scaling uses medians, means, and standard deviations from unique training months. Validation and test data use these same values.

A.3 Historical Market and Downside Exposures

This subsection defines the historical exposure controls used in Section 5.2. They are calculated from returns dated no later than formation month t. The estimates measure past market sensitivity and residual volatility; they are separate from the MDN's predicted mean and volatility.

Let $\mathcal{H}_t = \{t-59, \ldots, t\}$ be the 60 calendar months ending at formation. Within $\mathcal{H}_t$, let $\mathcal{P}_{i,t}$ contain the months with valid stock and market excess-return pairs. Here s indexes a historical month, $R_{i,s}$ is the raw stock return, and $R_{f,s}$ is that month's risk-free return. The stock excess return is $r^e_{i,s} = R_{i,s} - R_{f,s}$, and M^e_s is the market excess return. Estimate the following market regression with an intercept:

$$r^e_{i,s} = a_{i,t} + \beta_{i,t} M^e_s + u_{i,s}, \qquad s \in \mathcal{P}_{i,t}. \tag{44}$$

The intercept is $a_{i,t}$, the historical market beta is $\beta_{i,t}$, and $u_{i,s}$ is the historical regression disturbance. The exposure is indexed by t because it is re-estimated using the history available at each formation month. The regression requires at least 36 valid pairs and positive market-return variation. Idiosyncratic volatility is the residual standard deviation, calculated as

$$\sigma^{\text{idio}}_{i,t} = \left[\frac{\sum_{s \in \mathcal{P}_{i,t}} \widehat{u}^2_{i,s}}{|\mathcal{P}_{i,t}| - 2} \right]^{1/2}. \tag{45}$$

Here $\widehat{u}_{i,s}$ is the fitted residual, $\sigma^{\text{idio}}_{i,t}$ is idiosyncratic volatility, and the label idio means "idiosyncratic." The denominator subtracts two degrees of freedom for the estimated intercept and market slope. Define the downside subset $\mathcal{P}^-_{i,t}$ as

$$\mathcal{P}^-_{i,t} = \{s \in \mathcal{P}_{i,t} : M^e_s < \overline{M}^e_{\mathcal{H}_t}\}. \tag{46}$$

The superscript minus labels the downside subset. The threshold $\overline{M}^e_{\mathcal{H}_t}$ is the average market excess return over the full calendar window $\mathcal{H}_t$, including months when the stock return is missing. I estimate downside beta with the same intercept-inclusive regression on $\mathcal{P}^-_{i,t}$. It requires at least 12 downside pairs and at least 36 total pairs. All input data are dated no later than t. If the history is too short, the exposure remains missing and the observation is excluded from the common regression sample. It is not assigned zero beta.

A.4 Ordinary Controls and Monthly Transformations

Historical mean and volatility use $\{t-11, \ldots, t\}$ and require at least six valid returns. Momentum is $\exp\{\sum_{s=t-11}^{t-1} \log(1 + R_{i,s})\} - 1$, also with at least six valid observations.

Missing values for these ordinary history controls use the eligible-sample median in the formation month. This rule does not apply to betas. In the empirical regressions, explanatory variables are winsorized within month at 1% and 99%. Ordinary controls are then standardized. ES and contributions keep their winsorized decimal return values. These transformations apply only to the regressions; they do not change the saved neural forecasts.

B Model Implementation and Statistical Inference

B.1 Historical and Linear Benchmarks

Rolling Normal estimates each stock's mean and sample standard deviation from the 84 calendar months ending at formation. The mean requires at least one valid return and volatility at least two. If volatility is missing, I use the formation-month standard deviation across stocks. Standard deviations have a lower bound of 10^{-6}. For estimated mean $\widehat{\mu}$ and standard deviation $\widehat{\sigma}$, suppress the stock and formation-month indices. Write q_α for VaR and e_α for ES at tail probability α. The function Φ^{-1} is the inverse standard Normal CDF. These forecasts are

$$q_\alpha = \widehat{\mu} + \widehat{\sigma}\Phi^{-1}(\alpha), \qquad e_\alpha = \widehat{\mu} - \widehat{\sigma}\frac{\phi\{\Phi^{-1}(\alpha)\}}{\alpha}. \tag{47}$$

Linear Quantile estimates the 27 grid quantiles using firm variables, macro variables, and industry information. Each quantile is fitted with a month-weighted pinball loss. The predicted quantiles are sorted to prevent crossings. The reported tail moments and scores are calculated from this grid, with the endpoint quantiles held constant outside it. The model produces a set of quantiles, not an exact parametric density.

B.2 Fused Density Benchmarks

The fused benchmarks concatenate the 13 macroeconomic predictors m_t, the 21 firm predictors $x_{i,t}$, and a six-dimensional industry embedding $e_{i,t}$. Each model estimates its own embedding. Define the combined input $z_{i,t}^F = (m_t', x_{i,t}', e_{i,t}')' \in \mathbb{R}^{40}$, where the superscript F labels the fused model and primes denote transpose. A single network transforms these 40 inputs through hidden layers of widths 64, 32, and 8. Each hidden layer uses layer normalization and an exponential linear unit (ELU) activation; the first two also use dropout 0.05. Thus macroeconomic and firm information enter the same learned representation before any distributional parameters are produced.

Separate linear output heads map the final eight features to component means, scale scores, and, when there is more than one component, probability scores. Softplus transforms each scale score into a positive value, to which the model adds the same 0.005 scale floor used by the structured models. Softmax transforms the probability scores into weights that sum to one. Using the superscript F to distinguish the fused parameters, its predictive density is

$$f_{i,t}^{F}(r) = \sum_{k=1}^{K} \pi_{i,k,t}^{F} \mathcal{N}(r; \mu_{i,k,t}^{F}, (\sigma_{i,k,t}^{F})^2). \tag{48}$$

Here $\pi_{i,k,t}^{F}$, $\mu_{i,k,t}^{F}$, and $\sigma_{i,k,t}^{F}$ are the fused model's component probability, mean, and standard deviation. Fused K1 sets $K = 1$ and $\pi_{i,1,t}^{F} = 1$, giving a conditional Normal forecast. Fused K3 sets $K = 3$ and estimates all three probabilities. Because every output head receives the combined representation, macroeconomic and firm inputs can affect every parameter. In particular, Fused K3's probabilities can differ across firms in the same month; no common monthly latent state is imposed.

Both benchmarks minimize the marginal NLL, with equal total weight for each training month:

$$\mathcal{L}_{w}^{F} = -\frac{1}{|\mathcal{T}_w|} \sum_{t \in \mathcal{T}_w} \frac{1}{N_t} \sum_{i \in \mathcal{U}_t} \log f_{i,t}^{F}(R_{i,t+1}). \tag{49}$$

The symbol $\mathcal{L}_{w}^{F}$ is the fused training objective in window w. It has the same averaging rule as the marginal term in equation (14), evaluated with the fused density. Each stock–month is weighted inversely by its month's firm count, so months with more stocks do not receive more total weight. Training uses shuffled batches of 1,024 stock observations. Validation evaluates the same month-weighted marginal NLL and selects the checkpoint with the lowest value. These benchmarks have no joint-density loss term. They use the annual rolling samples, input scaling, and optimization settings described in this section, but their batches need not contain complete monthly cross-sections.

B.3 Structured Networks and Optimization

Figure 2 summarizes the network dimensions. The macro network has hidden layers of widths 16 and 8. The firm network has hidden layers of widths 64, 32, and 8. Its inputs combine the 21 scaled firm variables and the six-dimensional industry embedding. Hidden layers use layer normalization, exponential linear unit (ELU) activation functions, and dropout 0.05. Output layers produce component means and the values transformed by softplus into standard deviations. The FiLM layer converts the eight macro features into the eight-dimensional scale change $\Delta\gamma_t$ and shift β_t defined in equation (8). Both use the 0.1 scaling in that equation, and the lower bound on component standard deviations is

0.005.

The main models use AdamW with learning rate 0.003, weight decay 10^{-4}, gradient clipping at norm 1, and at most 500 epochs. Early stopping allows 20 epochs without a validation improvement of at least 10^{-6}. ReduceLROnPlateau multiplies the learning rate by 0.5 after five validation steps without improvement, down to a minimum of 10^{-6}. Structured training batches contain eight complete months. Months have equal total weight, no firms are subsampled, and each annual window starts with a new model. The main seed is 2027 plus the window index. The forecasts come from the 18 validation-selected models, without a final refit.

B.4 Calendar-Time Standard Errors

The forecast and portfolio tests use monthly loss differences or portfolio returns. The second-step contribution regressions use monthly estimated slopes. In each case, Newey–West standard errors allow dependence across months. When a test uses only bad months or only other months, I retain the original calendar gaps between observations. The calculation below makes this convention explicit.

For the monthly regression considered here, let x_t be its column vector of regressors, including the intercept, and X the matrix with rows x_t'. This local notation describes the monthly regression design, not a stock's neural input. Let $\widehat{u}_t$ be the fitted residual and $s_t = x_t\widehat{u}_t$ its regressor-weighted residual, often called a score. The covariance matrix of the estimated regression coefficients is

$$\widehat{V} = (X'X)^{-1}\widehat{\Omega}(X'X)^{-1}, \tag{50}$$

$$\widehat{\Omega} = \sum_t s_t s_t' + \sum_{h=1}^{L}\left(1 - \frac{h}{L+1}\right)\sum_t (s_t s_{t-h}' + s_{t-h}s_t'). \tag{51}$$

The matrix $\widehat{\Omega}$ sums contemporaneous score products and their lagged cross-products. The integer h is the calendar-month lag, L is the maximum lag, and $1-h/(L+1)$ is the Bartlett weight that decreases with lag length. Primes denote transpose and the exponent -1 denotes matrix inverse. The sums run over actual calendar positions; products involving an unavailable lagged month are zero. Months outside a selected evaluation group receive zero scores at their calendar positions. This keeps the original time gaps: two bad months a year apart are not treated as adjacent observations. The main lag length is $L = 12$. Reported p-values use the asymptotic Normal distribution and are not adjusted across hypotheses. Market rules and specifications were examined during the analysis; the reported tests evaluate each comparison separately.

C Additional Regression and Sorting Results

The additional tables retain the notation from the main text. A gap is a lowest-minus-highest group difference, and pp abbreviates percentage points. In Table C.3, R^2 is the coefficient of determination, the fraction of signal variation explained by its monthly linear projection; $100(1-R^2)$ is the residual percentage. Table C.4 uses "Beta gap" for the group difference in historical market beta and "Volatility gap" for the difference in predicted standard deviation.

Table C.1: Total ES and subsequent downside outcomes at both tail levels

Model	ES (%)	Outcome	All	p	Bad	p	Other	p
Structured K3	5	Crash	-0.540	< 0.001	-0.717	< 0.001	-0.495	< 0.001
Structured K3	10	Crash	-0.622	< 0.001	-0.823	< 0.001	-0.571	< 0.001
Structured K3	5	Excess loss	-0.075	< 0.001	-0.150	< 0.001	-0.056	< 0.001
Structured K3	10	Excess loss	-0.087	< 0.001	-0.173	< 0.001	-0.065	< 0.001
FiLM Sigma	5	Crash	-0.536	< 0.001	-0.685	< 0.001	-0.498	< 0.001
FiLM Sigma	10	Crash	-0.621	< 0.001	-0.792	< 0.001	-0.577	< 0.001
FiLM Sigma	5	Excess loss	-0.074	< 0.001	-0.139	< 0.001	-0.057	< 0.001
FiLM Sigma	10	Excess loss	-0.086	< 0.001	-0.161	< 0.001	-0.066	< 0.001
FiLM All	5	Crash	-0.542	< 0.001	-0.722	< 0.001	-0.496	< 0.001
FiLM All	10	Crash	-0.627	< 0.001	-0.831	< 0.001	-0.574	< 0.001
FiLM All	5	Excess loss	-0.073	< 0.001	-0.142	< 0.001	-0.055	< 0.001
FiLM All	10	Excess loss	-0.085	< 0.001	-0.165	< 0.001	-0.065	< 0.001

Notes: The first stage includes total ES, historical controls, and industry effects, omitting contribution. Entries are equal-month average ES slopes. A one-percentage-point increase in signed ES corresponds to the reported outcome change in percentage points. Both levels use the same history-complete sample. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

For Table C.3, I regress each signal on the other signal, all controls, and industry indicators. The coefficient of determination, R^2, is the fraction of within-month signal variation explained by this linear projection; the reported $100(1-R^2)$ is the percentage left in its residual. This R^2 is a goodness-of-fit statistic, not the square of an individual return. At 5% ES, the average share of variation left in the residual is about 7.5%–9.4% for contribution and 0.026%–0.036% for total ES. At 10% ES, the corresponding ranges are 5.8%–6.6% and 0.020%–0.026%. Total ES is therefore very closely related to the other regressors. The residual sorts complement the coefficient estimates by comparing stock groups formed from the remaining contribution variation. Table C.4 reports the groups' average characteristics.

Table C.2: Contribution slopes for returns and excess losses at both ES levels

Model	Outcome	δ_C: 5%	SE	p	δ_C: 10%	SE	p
Structured K3	Raw return	0.853	0.255	< 0.001	1.272	0.400	0.001
Structured K3	Excess loss	-0.317	0.142	0.026	-0.489	0.212	0.021
FiLM Sigma	Raw return	0.455	0.099	< 0.001	0.731	0.160	< 0.001
FiLM Sigma	Excess loss	-0.165	0.051	0.001	-0.274	0.086	0.002
FiLM All	Raw return	0.471	0.111	< 0.001	0.679	0.173	< 0.001
FiLM All	Excess loss	-0.194	0.068	0.004	-0.276	0.100	0.006

Notes: The full-control specification matches Table 12. Raw return and excess loss are unchanged across ES levels; excess loss is $\max\{-R - 0.10, 0\}$. Units are outcome percentage points per one-percentage-point increase in contribution. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

Table C.3: Remaining conditional signal variation at both ES levels

Model	Signal	Residual variance 5%	Residual variance 10%
Structured K3	ES	0.0259	0.0201
Structured K3	Contribution	7.5217	5.8007
FiLM Sigma	ES	0.0365	0.0264
FiLM Sigma	Contribution	9.4194	6.6043
FiLM All	ES	0.0340	0.0259
FiLM All	Contribution	8.8752	6.5739

Notes: Each signal is regressed on the other signal, the full controls, and industry effects separately by month and ES level. Entries are $100(1 - R^2)$ averaged equally across months. Entries measure the remaining linear variation.

Table C.4: Characteristics of residual contribution groups at both ES levels

Model	ES (%)	Market group	ES gap (pp)	Beta gap	Volatility gap (pp)
Structured K3	5	Bad months	-0.149	-0.034	0.064
Structured K3	5	Other months	0.041	-0.025	-0.031
Structured K3	10	Bad months	-0.197	-0.026	0.089
Structured K3	10	Other months	0.103	-0.027	-0.073
FiLM Sigma	5	Bad months	-0.252	-0.023	0.090
FiLM Sigma	5	Other months	0.019	-0.026	-0.037
FiLM Sigma	10	Bad months	-0.350	-0.015	0.154
FiLM Sigma	10	Other months	-0.103	-0.019	0.024
FiLM All	5	Bad months	0.364	-0.049	-0.213
FiLM All	5	Other months	-0.052	-0.024	-0.003
FiLM All	10	Bad months	0.007	-0.040	-0.051
FiLM All	10	Other months	-0.145	-0.020	0.055

Notes: Entries are equal-month average lowest-minus-highest residual decile differences. ES and predicted-volatility gaps are percentage points; market beta is in its original units. Residualization removes the monthly fitted linear relations; entries describe the characteristics of the resulting extreme groups.

D Market Definitions and Bandwidth Robustness

Table D.1: Alternative bad-market rules at both ES levels

Model	Market excess rule	Bad months	δ_C: 5%	p	δ_C: 10%	p
Structured K3	Bottom 20%	44	-4.114	< 0.001	-6.239	< 0.001
Structured K3	$\leq -2.5\%$	44	-4.114	< 0.001	-6.239	< 0.001
Structured K3	$\leq -3\%$	40	-4.181	< 0.001	-6.439	< 0.001
Structured K3	$\leq -5\%$	25	-2.706	0.002	-4.408	< 0.001
FiLM Sigma	Bottom 20%	44	-2.386	< 0.001	-3.767	< 0.001
FiLM Sigma	$\leq -2.5\%$	44	-2.386	< 0.001	-3.767	< 0.001
FiLM Sigma	$\leq -3\%$	40	-2.313	< 0.001	-3.689	< 0.001
FiLM Sigma	$\leq -5\%$	25	-1.797	< 0.001	-3.011	< 0.001
FiLM All	Bottom 20%	44	-2.343	< 0.001	-3.495	< 0.001
FiLM All	$\leq -2.5\%$	44	-2.343	< 0.001	-3.495	< 0.001
FiLM All	$\leq -3\%$	40	-2.287	< 0.001	-3.449	< 0.001
FiLM All	$\leq -5\%$	25	-1.873	< 0.001	-2.970	< 0.001

Notes: All specifications use full controls, the original signed contribution at the indicated ES level, and estimation-window effects. The individual stock crash cutoff remains -10%. Both levels use the same market dates under each rule. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

The alternative market rules use market excess returns. The individual stock crash cutoff

Table D.2: HAC bandwidth sensitivity at both ES levels

Model	Lags	δ_C: 5%	SE	p	δ_C: 10%	SE	p
Structured K3	3	-4.114	1.003	< 0.001	-6.239	1.556	< 0.001
Structured K3	6	-4.114	1.131	< 0.001	-6.239	1.745	< 0.001
Structured K3	12	-4.114	1.227	< 0.001	-6.239	1.883	< 0.001
FiLM Sigma	3	-2.386	0.362	< 0.001	-3.767	0.548	< 0.001
FiLM Sigma	6	-2.386	0.349	< 0.001	-3.767	0.529	< 0.001
FiLM Sigma	12	-2.386	0.345	< 0.001	-3.767	0.544	< 0.001
FiLM All	3	-2.343	0.335	< 0.001	-3.495	0.520	< 0.001
FiLM All	6	-2.343	0.347	< 0.001	-3.495	0.544	< 0.001
FiLM All	12	-2.343	0.363	< 0.001	-3.495	0.580	< 0.001

Notes: Point estimates use the same full-control monthly coefficients and the same 44 bad months at each tail level. Only the number of calendar-month Newey–West lags changes. Reported p-values are two-sided and unadjusted.

stays at a raw return below -10%. Using the 10% tail changes ES and its decomposition from the 5% version, but does not change this crash cutoff. These checks use the same full-control regression sample.

E Random Seeds and Training Protocols

E.1 Training Dynamics and Checkpoint Selection

The saved logs allow a comparison of validation dynamics across all 18 rolling windows. The fused models minimize marginal NLL. The structured specifications minimize the combination of normalized joint and marginal NLL in equation (16). To compare a common quantity, Figure E.1 uses validation marginal NLL for every model, including the marginal term recorded separately for the structured specifications. Each curve shows the median change from the end of epoch 1 across the 18 windows. All models have observations through epoch 20, so no curve is extended or filled after early stopping.

The structured specifications show continuing median validation improvement over this interval. Their validation trajectories also have smaller typical adjacent-epoch changes. For window w, define

$$V_w = \text{median}_{e=7,\ldots,20} \left| L_{w,e} - L_{w,e-1} \right|, \tag{52}$$

Here e is the training-epoch index, $L_{w,e}$ is validation marginal NLL in window w after epoch e, and V_w is the within-window variation summary. Absolute-value bars measure the magnitude of each adjacent-epoch difference, and median selects the middle value

of the resulting changes. This uses 14 successive changes within epochs 6–20, excluding the first five epochs from the variation summary while retaining them in the trajectory plot. A smaller V_w means smaller typical epoch-to-epoch changes in validation marginal NLL. In the main run, the median V_w is 0.02226 and 0.02357 for Fused K1 and Fused K3, compared with 0.00841, 0.00593, and 0.00785 for Structured K3, FiLM Sigma, and FiLM All. Table E.1 shows the same ordering between each structured specification and both fused benchmarks in each of the two additional validation seeds.

Table E.1: Validation dynamics and checkpoint selection across random seeds

Seed	Model	Epoch-1 checkpoints	Median checkpoint	Median V_w
2027	Fused K1	10/18	1.0	0.02226
2027	Fused K3	11/18	1.0	0.02357
2027	Structured K3	0/18	30.0	0.00841
2027	FiLM Sigma	0/18	26.0	0.00593
2027	FiLM All	0/18	21.5	0.00785
2026	Fused K1	14/18	1.0	0.03192
2026	Fused K3	7/18	3.0	0.02600
2026	Structured K3	0/18	22.0	0.00908
2026	FiLM Sigma	0/18	21.0	0.00923
2026	FiLM All	0/18	23.5	0.01044
42	Fused K1	7/18	4.0	0.02988
42	Fused K3	10/18	1.0	0.02409
42	Structured K3	0/18	20.0	0.00943
42	FiLM Sigma	0/18	21.5	0.00938
42	FiLM All	0/18	22.5	0.00925

Notes: Each row summarizes all 18 rolling estimation windows. The checkpoint is selected by marginal validation NLL for Fused and combined validation loss for the structured specifications. Epoch 1 is after one full training epoch, not initialization. For each window, $V_w = \text{median}_{e=7,\ldots,20} |L_{w,e} - L_{w,e-1}|$, where L is validation marginal NLL for every model; the last column is the median across windows. The statistic summarizes the magnitude of 14 adjacent changes. The fixed-40-epoch run has no separate validation trace. Objectives, batches, and update counts per epoch differ across the fused and structured procedures.

Checkpoint selection gives a related view. Fused K1 and Fused K3 retain epoch 1 in 10 and 11 windows in the main run. None of the structured specifications retains epoch 1 in any of the three validation seeds. The median retained epoch is 21.5–30 for the three structured specifications in the main run, compared with 1 for both fused models. Checkpoint selection uses each model's validation criterion.

Figure E.2 examines the two terms of the structured objective within each window. Both terms depend on the component probabilities, means, and scales; the joint term is not a separate loss for the probability network. The figure pairs their adjacent-epoch

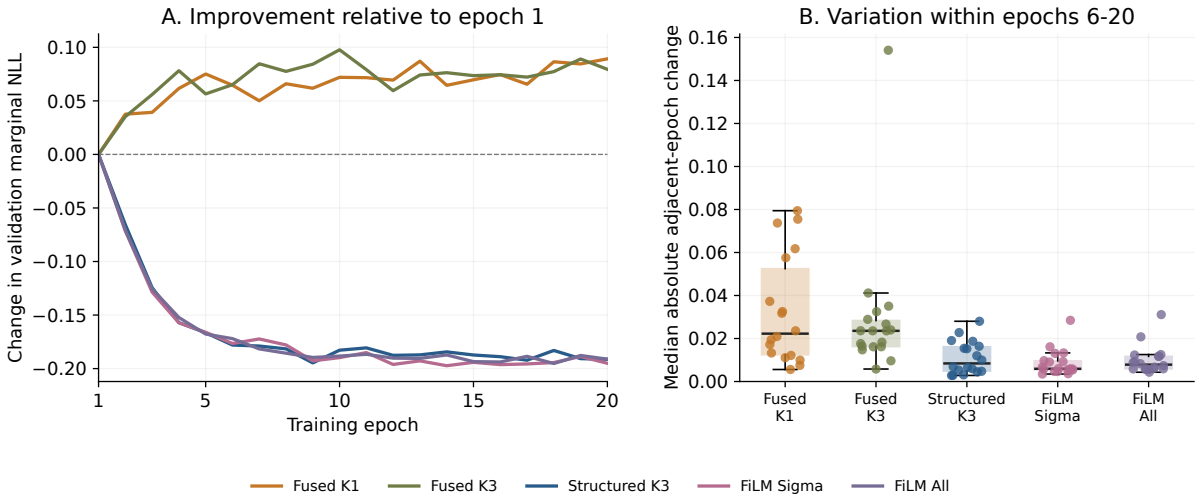


Figure E.1: Validation dynamics across the 18 rolling windows

Notes: The main run uses seed 2027. Panel A shows the across-window median of each window's change in validation marginal NLL relative to epoch 1, without smoothing. Negative changes indicate improvement. Panel B shows each window's V_w from equation (52); boxes summarize the distribution across windows and are not confidence intervals. The first epoch is a completed training epoch. All windows have at least 20 recorded epochs.

variation in training and validation. In the main run, the joint term varies less in 18, 16, and 13 training windows for Structured K3, FiLM Sigma, and FiLM All, respectively. The corresponding validation counts are 14, 13, and 10. Across all three validation seeds, the training comparison favors the joint term in at least 13 of 18 windows for each specification. The joint term has smaller adjacent-epoch changes more consistently in training than in validation. Training traces are accumulated with dropout active and parameters changing within the epoch, whereas validation traces evaluate the model after the epoch.

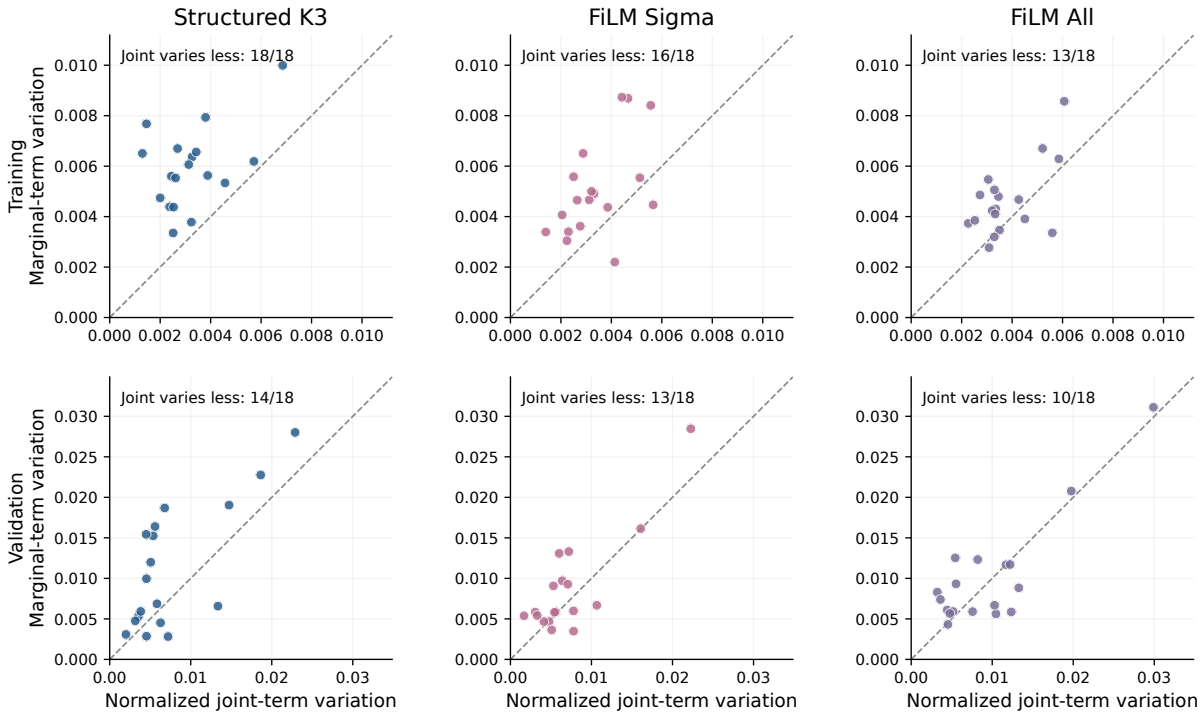


Figure E.2: Paired variation of the joint and marginal loss terms

Notes: Each point represents one of the 18 windows in the main run. Both coordinates use the median absolute adjacent-epoch change over epochs 6–20. The joint term is already normalized by the average training cross-section size. Above the dashed 45-degree line, the joint term varies less than the marginal term; below it, the reverse holds. Columns show the three structured specifications; rows show training and validation. Coordinate scales are common across models within a row.

These diagnostics support more sustained validation improvement and smaller typical validation changes under the structured training procedures. The fused procedure uses batches of 1,024 stock observations, while the shared-state procedures use batches of eight complete monthly cross-sections and a different loss. An epoch therefore also contains different numbers of parameter updates. These patterns complement the forecast evidence by showing how the structured models learn across rolling windows.

E.2 Random-Seed Sensitivity

Table E.2: Contribution and residual sorts across all four runs: 5% ES

Seed / protocol	Model	δ_C	p	Bad gap	p	Other gap	p
2027 / Val.	Structured K3	-4.114	< 0.001	6.449	< 0.001	0.091	0.836
2027 / Val.	FiLM Sigma	-2.386	< 0.001	7.141	< 0.001	-0.406	0.294
2027 / Val.	FiLM All	-2.343	< 0.001	6.588	< 0.001	0.093	0.806
2026 / Val.	Structured K3	-3.208	< 0.001	4.876	< 0.001	-0.572	0.156
2026 / Val.	FiLM Sigma	-1.931	< 0.001	4.553	< 0.001	-0.718	0.061
2026 / Val.	FiLM All	-1.483	< 0.001	3.581	< 0.001	-0.087	0.829
42 / Val.	Structured K3	-1.928	< 0.001	3.818	< 0.001	0.197	0.587
42 / Val.	FiLM Sigma	-1.589	< 0.001	4.208	< 0.001	0.066	0.890
42 / Val.	FiLM All	-1.730	< 0.001	4.559	< 0.001	0.201	0.618
2027 / 40	Structured K3	-4.620	< 0.001	5.678	< 0.001	0.489	0.290
2027 / 40	FiLM Sigma	-3.249	< 0.001	6.547	< 0.001	0.351	0.262
2027 / 40	FiLM All	-2.218	< 0.001	7.033	< 0.001	0.710	0.038

Notes: Full controls and the indicated ES level are used throughout. Gaps are lowest-minus-highest residual decile crash-rate differences, in percentage points. Val. denotes validation selection; 40 denotes fixed-40 training. All runs use the same outcomes and are evaluated separately without pooling. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

Tables E.2 and E.3 report contribution slopes and residual gaps at 5% and 10% ES for all four runs, including the fixed-epoch check discussed below. Seeds 2027, 2026, and 42 use the same validation procedure, inputs, and model settings. Seed 2027 is the main specification. Across the three validation runs and both ES levels, every bad-market residual gap is positive and significant at 1%. For 5% ES, all other-month gaps have $p > 0.05$. For 10% ES, the FiLM Sigma other-month gap with seed 2026 is negative with $p = 0.039$; the remaining eight have $p > 0.05$. The coefficient magnitudes vary across fits, so the recurring sign and residual sorting pattern are more useful for interpretation than a common numerical slope. Each run evaluates the same stock outcomes, with results reported separately.

Table E.3: Contribution and residual sorts across all four runs: 10% ES

Seed / protocol	Model	δ_C	p	Bad gap	p	Other gap	p
2027 / Val.	Structured K3	-6.239	< 0.001	6.463	< 0.001	-0.003	0.994
2027 / Val.	FiLM Sigma	-3.767	< 0.001	7.596	< 0.001	-0.205	0.624
2027 / Val.	FiLM All	-3.495	< 0.001	6.207	< 0.001	0.162	0.656
2026 / Val.	Structured K3	-5.112	< 0.001	5.146	< 0.001	-0.603	0.138
2026 / Val.	FiLM Sigma	-2.795	< 0.001	4.047	< 0.001	-0.835	0.039
2026 / Val.	FiLM All	-2.391	< 0.001	3.735	< 0.001	0.048	0.910
42 / Val.	Structured K3	-3.290	< 0.001	4.346	< 0.001	0.179	0.608
42 / Val.	FiLM Sigma	-2.314	< 0.001	3.946	0.001	-0.010	0.979
42 / Val.	FiLM All	-2.664	< 0.001	4.867	< 0.001	0.161	0.676
2027 / 40	Structured K3	-7.932	< 0.001	6.406	< 0.001	0.244	0.619
2027 / 40	FiLM Sigma	-4.971	< 0.001	6.659	< 0.001	0.057	0.847
2027 / 40	FiLM All	-3.476	< 0.001	6.888	< 0.001	0.463	0.194

Notes: Full controls and the indicated ES level are used throughout. Gaps are lowest-minus-highest residual decile crash-rate differences, in percentage points. Val. denotes validation selection; 40 denotes fixed-40 training. All runs use the same outcomes and are evaluated separately without pooling. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

Table E.4: Forecast scores across validation seeds at both tail levels

Seed	Model	CRPS	Tail CRPS 5%	Tail CRPS 10%	FZ0 5%	FZ0 10%
2027	Structured K3	0.072225	0.015872	0.026502	-1.364756	-1.563254
2027	FiLM Sigma	0.071225	0.015769	0.026332	-1.370090	-1.565342
2027	FiLM All	0.071661	0.015923	0.026527	-1.351581	-1.548862
2026	Structured K3	0.071721	0.016088	0.026698	-1.340979	-1.548172
2026	FiLM Sigma	0.071455	0.015922	0.026558	-1.360099	-1.556523
2026	FiLM All	0.071079	0.015943	0.026535	-1.355274	-1.555597
42	Structured K3	0.071052	0.015770	0.026256	-1.365504	-1.560350
42	FiLM Sigma	0.070675	0.015794	0.026289	-1.362932	-1.558994
42	FiLM All	0.071153	0.015981	0.026569	-1.349467	-1.543713

Notes: Scores are equal-month averages on each run's common seven-model sample. The primary seed is 2027. CRPS evaluates the whole distribution; tail CRPS and FZ0 use the indicated tail probability. Lower scores are preferred.

Table E.5: Neural forecast comparisons across all four runs and both tail levels

Seed / protocol	Model	NLL	CRPS	FZ0 5%	FZ0 10%
2027 / Val.	Fused K1	-0.50290	0.07803	-1.21732	-1.40765
2027 / Val.	Fused K3	-0.50514	0.08132	-1.17743	-1.36060
2027 / Val.	Structured K3	-0.64062	0.07222	-1.36476	-1.56325
2027 / Val.	FiLM Sigma	-0.66082	0.07123	-1.37009	-1.56534
2027 / Val.	FiLM All	-0.64746	0.07166	-1.35158	-1.54886
2026 / Val.	Fused K1	-0.44632	0.08123	-1.17600	-1.40285
2026 / Val.	Fused K3	-0.49974	0.08110	-1.03109	-1.32794
2026 / Val.	Structured K3	-0.64470	0.07172	-1.34098	-1.54817
2026 / Val.	FiLM Sigma	-0.65373	0.07146	-1.36010	-1.55652
2026 / Val.	FiLM All	-0.65848	0.07108	-1.35527	-1.55560
42 / Val.	Fused K1	-0.43570	0.08135	-1.05452	-1.31122
42 / Val.	Fused K3	-0.47375	0.08232	-1.04166	-1.30940
42 / Val.	Structured K3	-0.66042	0.07105	-1.36550	-1.56035
42 / Val.	FiLM Sigma	-0.67047	0.07068	-1.36293	-1.55899
42 / Val.	FiLM All	-0.65947	0.07115	-1.34947	-1.54371
2027 / 40	Fused K1	-0.40555	0.07778	-0.96724	-1.29369
2027 / 40	Fused K3	-0.50349	0.07956	-0.91684	-1.22740
2027 / 40	Structured K3	-0.62356	0.07350	-1.31337	-1.52002
2027 / 40	FiLM Sigma	-0.63970	0.07242	-1.32419	-1.53758
2027 / 40	FiLM All	-0.65375	0.07174	-1.34871	-1.55513

Notes: Each run uses its common seven-model forecast sample and its tail-level-specific common FZ0 validity masks. Lower losses are preferred. Val. denotes validation selection; 40 denotes fixed-40 training. Runs are reported separately.

Table E.5 compares each structured model with the fused benchmarks within each run. All three structured specifications have lower NLL, CRPS, and FZ0 at both the 5% and 10% levels than both benchmarks under every training configuration. The leading structured specification can change with the score and initialization. The common improvement across variants is the main result.

Table E.6: Overlap in extreme-ES stock selections across runs

Seed / protocol	ES tail (%)	Low: within	Low: vs. fused	High: within	High: vs. fused
2027 / Val.	5	88.2–90.7	67.7–77.6	91.2–93.7	72.5–74.7
2027 / Val.	10	88.0–90.6	66.6–77.2	90.8–93.4	69.7–73.2
2026 / Val.	5	90.2–90.8	71.0–75.7	91.4–93.0	71.3–73.2
2026 / Val.	10	90.1–90.8	68.6–74.8	91.2–92.8	68.9–71.6
42 / Val.	5	88.6–91.1	70.4–77.8	90.7–92.3	69.1–72.5
42 / Val.	10	88.5–91.1	68.9–77.2	90.5–92.1	67.3–70.7
2027 / 40	5	90.8–92.0	72.5–76.9	92.3–94.0	69.8–72.4
2027 / 40	10	90.7–91.8	71.0–76.2	91.8–93.7	66.9–69.9

Notes: Entries are ranges of equal-month average Dice overlap percentages. Within covers the three pairs among Structured K3, FiLM Sigma, and FiLM All; vs. fused covers their six comparisons with Fused K1 and Fused K3. Low and High refer to the lowest and highest ES stock deciles, each approximately 10% of the monthly cross-section, regardless of the ES tail level. Each run uses the common seven-model forecast universe of 480,446 stock–months over 216 months, before the additional portfolio eligibility restrictions. Val. denotes validation selection; 40 denotes fixed-40-epoch training. The table summarizes similarity in selected stock membership.

Table E.6 compares the range across the three structured–structured pairs with the range across the six structured–fused pairs. For each run, each ES tail level, and each extreme stock decile, the lowest within-family average overlap exceeds the highest structured–fused average overlap. The overlaps show that the structured variants select similar stocks on the same monthly universe.

E.3 Fixed-Epoch Training

The fixed-epoch results appear alongside the validation runs in Tables E.2, E.3, and E.5. The fixed-40-epoch run uses seed 2027 and combines training and validation into 84 months. It uses a constant learning rate, no validation checkpoint selection, and the final model at epoch 40. The architecture, predictor set, and test period are unchanged. This provides a check under an alternative training protocol.

The fully controlled contribution differences remain negative at both ES levels, with all six p-values below 0.001. The window-adjusted residual bad-minus-other crash gaps are 5.01, 6.40, and 6.89 percentage points at 5% ES, and 5.91, 6.86, and 6.97 points at

10% ES, also with $p < 0.001$. These results preserve the main conditional vulnerability pattern even though forecast scores and coefficient magnitudes change.

E.4 Shape and Portfolio Sensitivity

Table E.7: Full-distribution and fixed-cutoff shape comparisons across runs

Seed / protocol	Model	All NLL	p	Bad loss MSE	p
2027 / Val.	Structured K3	-0.011075	< 0.001	0.000010	0.010
2027 / Val.	FiLM Sigma	-0.007899	0.002	0.000014	0.017
2027 / Val.	FiLM All	-0.010257	0.009	0.000012	0.023
2026 / Val.	Structured K3	-0.012826	0.010	0.000008	0.036
2026 / Val.	FiLM Sigma	-0.009490	< 0.001	0.000018	0.028
2026 / Val.	FiLM All	-0.009690	< 0.001	0.000013	0.037
42 / Val.	Structured K3	-0.011819	0.003	0.000004	0.370
42 / Val.	FiLM Sigma	-0.009856	0.002	0.000009	0.068
42 / Val.	FiLM All	-0.005194	0.001	0.000007	0.238
2027 / 40	Structured K3	-0.013976	< 0.001	0.000001	0.845
2027 / 40	FiLM Sigma	-0.009411	< 0.001	0.000001	0.599
2027 / 40	FiLM All	-0.009073	0.005	0.000006	0.112

Notes: Differences are mixture minus its own moment-matched Normal. These metrics do not depend on the ES probability level. Excess loss uses the fixed raw-return cutoff of -10%. Negative differences favor the mixture. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

The matched-Normal comparisons retain significant full-sample NLL improvements for all three structured specifications in every run. Bad-market FZ0 improvements at both ES levels, and improvements in excess-loss forecasts, are less stable across scores and protocols. The full-sample NLL gain is the most consistent shape result across runs.

Tables E.9 and E.10 report total-ES portfolio returns and alphas at both tail levels for all four runs. Large positive bad-market spreads recur under both weighting schemes. Their full-period alphas vary with weighting and training configuration. The recurring bad-market return spreads show that total-ES rankings distinguish stocks with different realized losses during market declines.

Table E.8: Moment-matched Normal comparisons for both bad-market VaR–ES scores

Seed / protocol	Model	Bad FZ0 5%	p	Bad FZ0 10%	p
2027 / Val.	Structured K3	-0.000402	0.957	0.004539	0.153
2027 / Val.	FiLM Sigma	-0.004018	0.571	0.001888	0.575
2027 / Val.	FiLM All	-0.018730	0.056	-0.005685	0.284
2026 / Val.	Structured K3	-0.021523	0.091	0.002986	0.679
2026 / Val.	FiLM Sigma	-0.006245	0.220	0.010598	0.033
2026 / Val.	FiLM All	-0.013083	0.176	0.002021	0.693
42 / Val.	Structured K3	-0.015800	0.201	-0.003412	0.570
42 / Val.	FiLM Sigma	-0.012130	0.070	-0.000632	0.846
42 / Val.	FiLM All	-0.018478	0.126	-0.004266	0.464
2027 / 40	Structured K3	-0.032876	0.031	-0.004020	0.616
2027 / 40	FiLM Sigma	-0.029658	0.007	-0.008155	0.023
2027 / 40	FiLM All	-0.007348	0.319	0.003226	0.601

Notes: Differences are mixture score minus its own moment-matched Normal score in the same 44 bad months, using valid forecasts at each tail level. Each configuration is evaluated separately. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

Table E.9: ES portfolio results across all four runs: 5% ES

Seed / protocol	Model	Bad EW	Bad VW	Full alpha	p
2027 / Val.	Structured K3	6.184	5.887	0.750	0.006
2027 / Val.	FiLM Sigma	6.080	5.446	0.725	0.010
2027 / Val.	FiLM All	5.959	5.210	0.695	0.014
2026 / Val.	Structured K3	5.953	5.526	0.751	0.006
2026 / Val.	FiLM Sigma	6.048	5.651	0.748	0.005
2026 / Val.	FiLM All	5.735	5.168	0.637	0.023
42 / Val.	Structured K3	6.136	5.728	0.800	0.002
42 / Val.	FiLM Sigma	6.127	5.774	0.710	0.010
42 / Val.	FiLM All	6.058	5.349	0.732	0.008
2027 / 40	Structured K3	6.498	6.297	0.699	0.009
2027 / 40	FiLM Sigma	6.455	6.144	0.748	0.004
2027 / 40	FiLM All	6.474	6.154	0.683	0.009

Notes: Entries are monthly percentages for the highest-minus-lowest ES stock-decile portfolio. Bad EW/VW are mean returns in the same 44 bad months. Full alpha is the full-period equal-weighted FF5-plus-momentum alpha. The ES level is fixed within each table. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.

Table E.10: ES portfolio results across all four runs: 10% ES

Seed / protocol	Model	Bad EW	Bad VW	Full alpha	p
2027 / Val.	Structured K3	6.230	6.109	0.758	0.004
2027 / Val.	FiLM Sigma	6.097	5.516	0.726	0.008
2027 / Val.	FiLM All	5.905	5.112	0.673	0.017
2026 / Val.	Structured K3	5.970	5.446	0.751	0.007
2026 / Val.	FiLM Sigma	6.071	5.741	0.759	0.004
2026 / Val.	FiLM All	5.756	5.193	0.652	0.019
42 / Val.	Structured K3	6.125	5.678	0.796	0.002
42 / Val.	FiLM Sigma	6.142	5.720	0.717	0.009
42 / Val.	FiLM All	6.082	5.388	0.727	0.009
2027 / 40	Structured K3	6.428	6.277	0.704	0.008
2027 / 40	FiLM Sigma	6.397	6.013	0.732	0.005
2027 / 40	FiLM All	6.487	6.288	0.677	0.009

Notes: Entries are monthly percentages for the highest-minus-lowest ES stock-decile portfolio. Bad EW/VW are mean returns in the same 44 bad months. Full alpha is the full-period equal-weighted FF5-plus-momentum alpha. The ES level is fixed within each table. Months receive equal weight. Two-sided unadjusted p-values use NW12 standard errors.